Contents

STRATO v2.0 Phase 2 pseudocode: decay, source, inventory, Arrhenius diffusivity and temperature histories **1**
1 Reading guide . . . 1
2 Overview of one v2.0 run . . . 3
3 One time step, side by side: v1.0 and v2.0 . . . 5
4 Blocks in run order . . . 6
5 Reference solutions (outside the finite-volume loop) . . . 22
6 Glossary of variables . . . 31
7 Verification cases and pass criteria . . . 35
8 Cross-check of every step against the reconciled equations . . . 39
9 Status, open items and decisions . . . 42
10 Evidence . . . 51
References . . . 52

STRATO v2.0 Phase 2 pseudocode: decay, source, inventory, Arrhenius diffusivity and temperature histories

Date: 2026-11-06; open items updated 2026-11-13 (Sections 9.2, 9.3 and 9.5). Status: DRAFT FOR REVIEW.

Not implemented yet: Rust follows after review. No Rust has been written for Phase 2. No file of STRATO v1.0.0 under `src/`, `tests/` or `examples/` has been changed. This page is the plain-language pseudocode that comes after the mathematics (Sections 3 to 5 of the derivation, equations (P21.1) to (P21.92) and (P22.1) to (P22.59)) and before any code. Work stops here until the review is finished. The open-item work of 2026-11-13 added the derivation equations (P21.93) to (P21.142) and (P22.60) to (P22.142); this page cites them in notes and in the proposals of Section 9.5, and no algorithm step uses them yet.

1 Reading guide

1.1 What this page is

Phase 2 of STRATO adds two things to the v1.0 finite-volume solver (Code B) and to the modal series (Code C).

- Phase 2.1: radioactive decay, a fission-product source and a full inventory account (generated, decayed, released, still inside, still undecayed outside).
- Phase 2.2: a diffusivity that follows temperature by an Arrhenius law, and a temperature that follows a history in time.

The two parts were first written as separate drafts: “the decay part” (Phase 2.1, blocks A to F) and “the temperature part” (Phase 2.2, blocks T1 to T7). Their block codes appear in this page only in the lines that say where a block came from and in Section 8. This page joins them into one run of the program, in the order in which the program does things, and then checks every step against the equation it cites. Section 8 lists the places where the two drafts disagreed or where an equation was cited loosely. In each case the pseudocode was changed. No equation was changed.

1.2 Step format

Each block is a list of numbered short steps. A step has a code such as H4.2 and a plain-language action. A line that begins with “why:” gives the reason. A step that rests on an equation names it in parentheses, for example (P21.58). A step with no equation number is bookkeeping, an input check or a design rule.

Three marks show how a step relates to v1.0.

Table 1.1 Change marks.

Mark	Meaning
[KEPT]	the step is identical to STRATO v1.0.0
[CHANGED]	a v1.0.0 step gets a different formula
[NEW]	the step does not exist in v1.0.0

Four words show how firm the mathematics behind a step is.

Table 1.2 Status words for steps.

Word	Meaning
APPROVED	the equation is in the reconciled Sections 3 to 5 and was re-derived in B3 or covered by the section check logs and scripts
DESIGN	a choice taken in this pseudocode that no equation fixes (a default, a data layout, a safety rule); the review may overrule it
OPEN	the equation or its input data is not settled; the step says what is missing
HELPER	a small identity used only to arrange a formula for stable arithmetic; proved in a check script and not part of the reconciled sections

Every equation or operation is also classified as IMPLEMENTED (with file and line in the v1.0.0 tag), DERIVED NOT IMPLEMENTED (derived and checked, no Rust yet) or FUTURE (outside Phase 2, documented only). Every numerical property is classified as PROVED, DEMONSTRATED NUMERICALLY or CLAIMED ONLY. Section 9 gives the tables.

1.3 Equation numbers and names

Equations (P21.x) come from Sections 3 and 4 and equations (P22.x) from Section 5. Three small identities for stable arithmetic carry the numbers (H21.1) to (H21.3) and two design relations carry (H22.1) and (H22.2); they appear where they are used. Code links point to the tag v1.0.0, for example src/numerical/finite_volume.rs, lines 162 to 182 is the v1.0 Backward Euler step.

Units inside the program are SI: metres, seconds, kelvin, joules per mole, moles per cubic metre. A number that enters in other units carries a unit tag and is converted first (block A). Names in lower case with underscores are values. Names with a leading capital (`Law`, `History`) are records, that is, labelled bundles of numbers. A list is written `[a, b, c]` and is indexed from 1 in the text and from 0 in the planned code.

The two drafts used some symbols twice. The pseudocode uses the names in Table 1.3 so that one name means one thing. Section 6 gives the full glossary.

Table 1.3 Renamed symbols.

Name in this page	Replaces	Reason
`gen`, `gen_rate`	G (generation rate in mol s^{-1})	G is also the face conductance of the finite-volume step
`west_g`, `east_g`, `G_R`	g_j^W, g_j^E, G_R	unchanged; these are the conductances
`mode_kappa`	$\kappa_n = \Lambda_n + \lambda$	`ratio_i` now names the layer ratio of Eq. (H22.2), formerly also κ_i
`Rg`	R in the Arrhenius code	R is the outer radius of the particle
`t_event`	e for an event time	`pivot_e[j]` keeps e_j of Eq. (P21.61)
`arg_mode`, `arg_branch`	x	x was used for three different arguments

2 Overview of one v2.0 run

2.1 Flow of one run

```
RUN FILE (regions, laws, nuclide, source, temperature history, time settings)
     |
     v
[A] check the input, convert units to SI
     |
     +--> [B] decay constant        (from the half-life)
     +--> [C] diffusivity laws      (D0 and Q of each layer)
     +--> [D] temperature history   (T as a function of time)
     +--> [E] source history        (fission rate, yield, source per interval)
     |
     v
[G] geometry once                   (volumes, face areas, distances, cell factors)
[F] list of events                  (breakpoints, source switches, output times)
     |
     v
+-------------------------- TIME LOOP, one pass per step ---------------------------+
|  [F] choose the next step: t_old --> t_new, dt, landing on every event            |
|  [H1] diffusivity of each layer over this step                                    |
|  [H2] conductances from those diffusivities                                       |
|  [H3] source of each cell, averaged over the step                                 |
|  [H4] one Backward Euler solve with decay (v1.0 elimination, new diagonal)        |
|  [J]  bookkeeping: inside, generated, decayed, released, outside, residuals       |
|  [K]  write a row if t_new is an output time                                      |
+-----------------------------------------------------------------------------------+
     |
     v
[K] end-of-run report (closure, positivity, warnings, counts)

Reference solutions, outside the loop and used to test it:
[R1] modal series with decay and source   [R2] effective time (shared scaling only)
[R3] piecewise-constant temperature       [R4] staircase of a ramp (test tool)
```

Read the diagram from top to bottom. The set-up blocks run once. The loop runs once per time step. Blocks R1 to R4 never run inside a finite-volume run; they produce reference numbers to compare with it.

2.2 Blocks in run order

Table 2.1 Blocks in run order.

Block	Content	Origin in the part drafts	Main equations	Planned Rust module
A	input and unit checks	new glue; T and Q unit rules from T1	(P22.10)	`src/config.rs` (extended)
B	decay constant from half-life	decay part, block A	(P21.5) to (P21.7)	`src/physics/decay.rs`
C	Arrhenius diffusivity of a layer	temperature part, T1	(P22.1), (P22.5) to (P22.7), (P22.9)	`src/physics/diffusivity.rs`
D	temperature history	temperature part, T2	(P22.21), (P22.39), (P22.58), (P22.59)	`src/thermal/temperature_history.rs`
E	yield and source history	decay part, block B	(P21.8), (P21.10) to (P21.12), (P21.52), (P21.57)	`src/physics/fission_source.rs`
F	events and step planner	decay part C1 and temperature part T3, merged	(P21.77) to (P21.79), (P22.19), (P22.20)	`src/numerical/finite_volume.rs` (planner) and `src/thermal/temperature_history.rs`
G	geometry once, cell factors	temperature part T4.1 and decay part B3	(P21.54), (P22.14)	`src/numerical/finite_volume.rs` (extended)
H	one time step	T4.2, T4.3, B4, B5, C2	(P21.57) to (P21.62), (P22.13) to (P22.17), (P22.22)	`src/numerical/finite_volume.rs` (extended)
J	bookkeeping	decay part D and T4.5	(P21.40) to (P21.45), (P21.70) to (P21.76), (P22.16)	`src/physics/inventory.rs`
K	output rows and end report	decay part D9, D10	(P21.45)	`src/physics/inventory.rs`, `src/main.rs`
M	the main loop that calls A to K	new glue	none	`src/numerical/finite_volume.rs` (run extended)
R1	modal series with decay and source	decay part, block E	(P21.24) to (P21.49)	`src/analytical/modal_source.rs`
R2	effective time	temperature part T5	(P22.24) to (P22.37)	`src/analytical/thermal_intervals.rs`, `src/thermal/temperature_history.rs`
R3	piecewise-constant temperature with projection	temperature part T6	(P22.39) to (P22.46)	`src/analytical/thermal_intervals.rs`, `src/analytical/modal_source.rs`
R4	staircase of a ramp (test tool)	temperature part T7	(P22.53) to (P22.57)	`src/analytical/thermal_intervals.rs`

2.3 Which solution path serves which input

Table 2.2 Solution paths.

Input	Path	Notes
any allowed input	finite volume, blocks A to M	the production path; first order in time, second order in space
constant diffusivities, uniform kernel start and kernel source, absorbing surface	R1	exact in time for a piecewise-constant source
one layer, or all layers sharing one temperature dependence, any decay, any source	R2, after the structural test of step R2.1	refused with a reason when the test fails; real TECDOC-978 data fail it for five layers
temperature constant on each interval, layers do not share a dependence	R3	exact for the untruncated series; the reference for a temperature step such as CRP-6 case 3c
temperature ramp, layers do not share a dependence	R4 (test tool) or the finite-volume path	no general closed form is available and no effective time exists (Section 5.4.2 of the derivation); R4 is slow and serves verification only

2.4 Planned Rust modules (proposal)

None of these exists yet. The v1.0 files stay as released until the review is finished.

Table 2.3 Planned modules.

Planned module	New or extended	Blocks
`src/physics/decay.rs`	new	B
`src/physics/fission_source.rs`	new	E, and the cell factor of G
`src/physics/inventory.rs`	new	J, K
`src/physics/diffusivity.rs`	new	C, and the step average of H1
`src/thermal/temperature_history.rs`	new	D, the temperature part of F, and R2
`src/numerical/finite_volume.rs`	extended	planner F, G, H, M
`src/analytical/modal_source.rs`	new	R1, and the propagation of R3
`src/analytical/thermal_intervals.rs`	new	R2, R3, R4
`src/config.rs`	extended	A
`tests/`, `examples/verify.rs`	new files and extended	Section 7

3 One time step, side by side: v1.0 and v2.0

The left column follows the v1.0.0 code: build_mesh, lines 50 to 99, the solver, lines 132 to 160, the Backward Euler step, lines 162 to 182 and the run loop, lines 184 to 236. The right column is the plan of this page. The line numbers were read from the tag.

Table 3.1 The v1.0 time-loop step and the v2.0 step.

Item	v1.0 FV time-loop step	v2.0 step
mesh and conductances	`build_mesh` fixes volumes, centroids and series conductances once, from one constant diffusivity per cell (lines 78 to 91)	block G fixes everything that does not depend on D once; block H2 recomputes the conductances in every step from the layer diffusivities of that step (P22.14) [CHANGED]

Item	v1.0 FV time-loop step	v2.0 step
initial state	kernel cells hold the initial concentration, all others zero (lines 114 to 120 sum the inventory)	same; all zero is allowed when a source is present [CHANGED]
start check	stop if the initial inventory is not positive (lines 186 to 189)	stop only if the initial inventory is zero and the source is zero everywhere (J0) [CHANGED]
time grid	`steps = ceil(end_time / dt)`, `dt = end_time / steps`, one constant step (lines 190 and 191)	FIXED mode keeps this rule bit for bit; PLANNED mode lands on every event and respects the decay, diffusion and temperature limits (block F) [NEW]
diagonal excess of the matrix	`v[i] = volume / dt` (line 178)	`volume / dt + decay_constant * volume` (H4.2) (P21.59) [CHANGED]
right-hand side	`rhs[i] = v[i] * old[i]` (line 179)	`(volume / dt) * old[i] + volume * source_cell[i]` (H4.2) (P21.59) [CHANGED]
elimination	subtraction-free tridiagonal solve (lines 132 to 160)	unchanged (H4.3) (P21.61) [KEPT]
release rate	`mesh.outer_conductance * next[last]` (line 210)	`G_R * conc_new[N]` with the G_R of this step (J1.2) (P22.16) [CHANGED]
release sum	Kahan sum of `dt * release_rate` (lines 211 to 217)	Kahan sums of the release, the generated amount, the decayed amount and the decayed-outside amount (J1.3) (P21.72) [CHANGED]
inventory	`inventory(mesh, c)` (line 219)	same function (J1.1) [KEPT]
residual	`inv + released - initial` (line 220)	two residuals, inside and outside, relative to $M_0 + M_{\text{gen}}$ (J1.5) (P21.72), (P21.75) [CHANGED]
atoms outside the particle	not tracked	`out_now` and the decayed-outside sum (J1.4) (P21.73), (P21.75) [NEW]
output rule	every `output_every_steps` steps and the last step (line 221)	at the output times (PLANNED) or every `output_every_steps` steps (FIXED) [CHANGED]
positivity	reported as the minimum concentration; the v1.0 unit test accepts -10^{-13} (line 310)	reported; the proof says it is not below 0.0, so any negative value is a failure (J1.6) (P21.62), (P22.17) [CHANGED]
temperature	none; D is constant	block D supplies $T(t)$; block C turns it into D [NEW]
decay	none	λ from block B enters only the diagonal [NEW]
source	none	block E and H3 [NEW]

4 Blocks in run order

4.1 Block M: the main loop

Block M only calls the other blocks in the right order. It adds no mathematics.

```
BLOCK M  one complete run                        planned module: src/numerical/finite_volume.rs (run,
       ↪  extended)
INPUT   the checked run description of block A
OUTPUT  rows at the output times; the final concentration profile; the end-of-run report

M1  Set-up, once per run, in this order:
      decay_constant, mean_life_s       = block B
      laws, settings                    = block C
      history                           = block D          (none for a constant-temperature run)
      yield, source_history             = block E
      geometry, cell_factor             = block G
      events, step caps                 = block F, steps F1 to F4
M2  Start state: conc[j] = initial kernel concentration in the kernel cells, 0 in all others.
    Run block J, step J0 (totals set to zero, inv0 computed).
    Write the row for t = 0.
M3  WHILE the last event (t_end) has not been reached:
      (t_old, t_new, dt, flags)         = block F, step F5
      D_layer[1..5]                     = block H, step H1
```

```
      west_g[j], east_g[j], G_R         = block H, step H2
      source_cell[j], gen_rate          = block H, step H3
      conc_new[j]                       = block H, step H4
      block J, step J1, with conc_new, G_R, gen_rate and dt
      IF t_new is an output time THEN write a row (block K, step K1)
      conc = conc_new
M4  Write the end-of-run report (block K, step K2).
    why: every quantity that enters the matrix, the source integral and the bookkeeping of one
         step is computed from the same t_old, t_new, dt and G_R, so the balance can close.
```

4.2 Block A: input and unit checks

Origin and status. New glue. The rules for tags come from the temperature draft (steps T1.1 and T1.2). Equation used: (P22.10) for $T = T_C + 273.15$ K, APPROVED.

```
BLOCK A  input and unit checks                        planned module: src/config.rs (extended)
INPUT   the run file of v1.0 plus the keys named below
OUTPUT  a checked run description in SI units, or an error that names the field   [NEW]

A1  Regions: exactly five, outer radii strictly increasing, at least one cell each.            [KEPT]
    why: this is the v1.0 rule (src/config.rs lines 79 to 82). Decision 1 asks whether to
         allow a one-region particle for the Booth test cases.
A2  Nuclide: a name (block B reads its half-life) or "none" (decay_constant = 0).
A3  Source: a fissile mix with yields (block E) and a fission rate history, or "none".
A4  Layer laws: for each layer either a constant diffusivity (stored as one branch with Q = 0)
    or a list of branches with D0 and Q, with the unit tag of Q.
A5  Temperature history: kind LINEAR or CONSTANT with unit tags (block D), or "none".
A6  Time settings: t_end, grid mode FIXED or PLANNED, the request dt (FIXED) or the planner
    settings (PLANNED), output times or output_every_steps, step limit.
A7  Initial kernel concentration in mol per m3, nonnegative.
A8  Convert every value to SI using its unit tag. A missing or unknown tag is an error that names the
       ↪ field.
    degC to K: T = T_C + 273.15.                                                (P22.10) APPROVED
    why: a missing tag would turn 1600 degC into 1600 K, or kJ per mol into J per mol.
A9  Consistency:
      IF any branch has Q > 0 THEN a temperature history must be given (it may be one constant interval).
      IF grid mode is FIXED THEN there must be no source switch time, no temperature knot and no output
        ↪ time list.
      IF the nuclide is an activation product THEN stop (block E, step E1).
    why: the fixed grid cannot land on an event; the planner can.
```

4.3 Block B: decay constant from half-life

Origin. Decay part, block A.

Table 4.1 Equations of block B.

Equation	Statement	Status	Class
(P21.5), (P21.6)	$dN/dt = -\lambda N$, $N = N_0 e^{-\lambda t}$	APPROVED	DERIVED NOT IMPLEMENTED
(P21.7)	$\lambda = \ln 2 / t_{1/2}$, mean life $1/\lambda$	APPROVED	DERIVED NOT IMPLEMENTED

The nuclear data table `data/nuclear/decay.csv` carries, per nuclide, the half-life as written in ENSDF (number and unit), the half-life in seconds and λ to seven digits. Its year follows the IAEA LiveChart convention, 1 y = 31 556 925.97 s. A calendar year of 365.25 d (31 557 600 s) would change λ by 2.1×10^{-5} relative, so the unit table below must not be replaced by a rounder value. The half-lives are the ENSDF adopted values, not the half-lives printed in TECDOC-1674 Table 8.1.

```
BLOCK B  decay constant                               planned module: src/physics/decay.rs
INPUT   nuclide name; half-life number and unit as in the nuclear data table
OUTPUT  decay_constant (1/s); mean_life_s (s)   [NEW]
```

```
B1  IF the half-life field is the word STABLE THEN
        decay_constant = 0 ; mean_life_s = "none" ; leave the block.
    why: a stable nuclide has no loss term. Blocks H and J then reduce to the v1.0 equations
         exactly, which is the regression test V1.
B2  Look up seconds_per_unit: s = 1, minute = 60, hour = 3600, day = 86400, year = 31 556 925.97.
    IF the unit is not in this list THEN stop with an error that names the unit.
    why: a silent default would give a wrong lambda by a factor such as 60.
B3  half_life_s = half_life_number * seconds_per_unit.
B4  IF half_life_s is not finite or not greater than 0 THEN stop with an error.
B5  decay_constant = ln(2) / half_life_s.                                    (P21.7) APPROVED
    why: setting N(t_half) = N0/2 in N = N0 exp(-lambda t) gives exp(-lambda t_half) = 1/2.
B6  mean_life_s = 1 / decay_constant.                                        (P21.7) APPROVED
B7  Consistency checks (report a warning, do not change the value):
      (a) decay_constant * half_life_s equals ln(2) to 1e-12 relative (a dimensionless product;
          any other result means a unit slip).
      (b) IF the table also gives half_life_s, B3 agrees with it to 1e-8 relative.
      (c) IF the table also gives lambda, decay_constant agrees with it to 5e-7 relative
          (the column has seven printed digits).
```

Table 4.2 Decay constants that block B must reproduce (recomputed from `decay.csv` by `b4_phase2_pseudocode_integration.py`).

Nuclide	Half-life (ENSDF)	$t_{1/2}$ in s	λ in s^{-1}	Mean life in s
Cs-137	30.08 y	9.492323×10^{8}	7.302187×10^{-10}	1.3695×10^{9}
Sr-90	28.91 y	9.123107×10^{8}	7.597709×10^{-10}	1.3162×10^{9}
Ag-110m	249.83 d	2.158531×10^{7}	3.211198×10^{-8}	3.1141×10^{7}
I-131	8.0252 d	6.933773×10^{5}	9.996681×10^{-7}	1.0003×10^{6}

4.4 Block C: Arrhenius diffusivity of a layer

Origin. Temperature part, T1.

In plain words. A diffusivity that follows temperature is stored as a short list of terms. Each term is a pair, a pre-exponential factor D_0 and an activation energy Q. At a given temperature each term gives $D_0 e^{-Q/(R_g T)}$ and the layer diffusivity is the sum of its terms, because two parallel transport paths add their fluxes. A layer that keeps the constant diffusivity of v1.0 is the special case of one term with $Q = 0$.

The program never forms $e^{-Q/(R_g T)}$ alone for a large Q, because it can underflow to exactly zero: for the Kr and Xe law in PyC ($D_0 = 2.0 \times 10^{5}$ m^2 s^{-1}, $Q = 923$ kJ mol^{-1}, IAEA, 1997, Table A-2) at 100 K the logarithm of the term is -1097.9 while the exponential is 0 in double precision. It works with logarithms and adds the terms with the largest one factored out.

Table 4.3 Equations of block C.

Equation	Statement	Status	Class
(P22.1)	$D_i(T) = \sum_m D_{0,i,m} \exp(-Q_{i,m}/(R_g T))$	APPROVED	DERIVED NOT IMPLEMENTED
(P22.5)	$R_g = N_A k_B$	APPROVED	DERIVED NOT IMPLEMENTED
(P22.6)	apparent activation energy Q_{eff} as the D-weighted mean of the branch energies	APPROVED	DERIVED NOT IMPLEMENTED
(P22.7)	crossover temperature T_x of a two-branch law	APPROVED	DERIVED NOT IMPLEMENTED
(P22.9)	fluence factor $\exp(\Phi_f/5)$ on one SiC branch, Φ_f in 10^{25} m^{-2} with cutoff $E > 16$ fJ	formula APPROVED; conversion of the fluence cutoff OPEN (data)	DERIVED NOT IMPLEMENTED

Equation	Statement	Status	Class
(P22.10)	$T = T_C + 273.15$ K	APPROVED	DERIVED NOT IMPLEMENTED

```
Branch   = { D0 [m^2/s], Q [J/mol] }
Law      = { branches : list of Branch (length >= 1; the data use 1 or 2),
             fluence  : optional { branch_index, scale_1e25_per_m2 = 5.0, cutoff_tag } }
Window   = { T_min [K], T_max [K] }     -- optional validity range, only if printed in the source
Settings = { gas_constant : SI | FIXED_8_3143,   -- default SI
             D_floor [m^2/s] > 0 }
```

```
BLOCK C  Arrhenius diffusivity                        planned module: src/physics/diffusivity.rs
INPUT   Law of each layer; temperature T in K; Settings; frozen fluence Phi_f (0 if none)
OUTPUT  D in m^2/s; apparent activation energy in J/mol                                  [NEW]

C1  load_law(raw_branches, q_unit, window, fluence)
      1. require q_unit in {"J/mol", "kJ/mol"}; otherwise stop with an error naming the field.
         why: TECDOC-978 defines Q in J/mol in the text but prints kJ/mol in every table column;
              a missing tag would be a silent factor 1000.
      2. for each branch: require D0 finite and D0 > 0; require Q finite and Q >= 0.
         why: ln(D0) must exist. Q = 0 is a real row (iodine in matrix pores), so zero is allowed.
      3. convert Q to J/mol (multiply by 1000 if the tag was kJ/mol).
      4. IF fluence is given THEN require cutoff_tag == "E>16fJ" (0.1 MeV).        (P22.9) [OPEN: data]
         why: AGR-1 fluences are quoted for E > 0.18 MeV; the conversion factor between the two
              cutoffs is NOT VERIFIED and must not be guessed. Any other tag is an error until a
              verified conversion exists.
      5. store the window if given. Never store a default window.

C2  ln_branch(law, m, T, settings, Phi_f)
      1. D0_eff = D0[m]; IF the fluence rule applies to branch m THEN D0_eff = D0[m] * exp(Phi_f /
        ↪  scale).   (P22.9)
         why: Phi_f is frozen during a post-irradiation heating test, so the factor is a constant
              of the run and not a function of time.
      2. Rg = 8.31446261815324 if settings.gas_constant == SI, else 8.3143.          (P22.5)
      3. return ln(D0_eff) - Q[m] / (Rg * T).                                        (P22.1)

C3  diffusivity(law, T, settings, Phi_f)
      1. IF the law has one branch and Q == 0 THEN return D0 exactly (no exp, no log).   [DESIGN]
         why: a constant layer then reproduces the v1.0 number bit for bit (regression test VT16).
      2. logs = [ln_branch(law, m, T, ...) for every branch m]
      3. top = maximum of logs
      4. D = exp(top) * sum over m of exp(logs[m] - top)                             (P22.1)
         why: log-sum-exp keeps every term finite; the largest term is exactly 1.
      5. IF a window exists and T is outside it THEN record a warning (once per run, with layer
         and temperature range). Do not clamp and do not stop.
      6. IF D < D_floor THEN D = D_floor and count one floor event; report the count at the end.
         why: the conductance divides by D, and the series uses sqrt(Lambda / D).
      7. return D.

C4  apparent_activation_energy(law, T, settings)
      1. logs as in C3 step 2; top = maximum; w[m] = exp(logs[m] - top).
      2. return sum over m of Q[m] * w[m] / sum over m of w[m].                      (P22.6)
         why: this is the D-weighted mean of the branch energies, the slope of ln D against 1/T.
              It moves from Q_1 to Q_2 as T rises, and it sets the step limit of block F.

C5  crossover(law)      -- diagnostic only, two-branch laws
      return (Q[2] - Q[1]) / (Rg * ln(D0[2] / D0[1])).                               (P22.7)
```

Checks already made on these steps (Python). Bad inputs (D0 equal to zero, D0 negative, Q negative, Q not a number, missing Q tag, $T = 0$ K, $T = -300$ °C, missing temperature tag) are rejected and $Q = 0$ is accepted (`b4_arrhenius_checks.py`). $D(T)$ by the log-sum-exp rule agrees with 40-digit arithmetic for four register laws at five temperatures, worst relative difference 8.4×10^{-15}. For Cs in SiC (FRG, $\Phi_f = 0$, register rows A3-01 and A3-02) the crossover is 1772.4 K and Q_{eff} is 130.10 and 438.22 kJ mol^{-1} at 1250 and 1600 °C.

Note on the gas constant (DESIGN, decision 11). The register stores the R_g used by each source row (8.3143 for the TECDOC-978 rows, 8.3145 for the Collin et

al. (2016) and IAEA (2012) rows; the BISON rows from Hales et al. (2021) print no value, while Collin et al. (2021) give 8.3145 for BISON, register RG-5). The default is the SI value for every row, with `FIXED_8_3143` as the option for four-digit comparison with TECDOC-1674. CRP-6 case 1a (bare kernel, $r = 250$ μm, 200 h at 1200 °C, two-branch Cs in UO_2) gives $F = 0.467337$ with 8.3143 and $F = 0.467404$ with the SI value; only the first rounds to the printed 0.4673. Which constant the CRP-6 participants used is NOT VERIFIED.

4.5 Block D: temperature history

Origin. Temperature part, T2, with one new step (D6) found in the cross-check of Section 8.

In plain words. The history says what temperature the whole particle has at each moment (assumption A1: the temperature is uniform in the particle). Two kinds are needed.

- **Linear kind.** A table of breakpoints (t_k, T_k) with strictly increasing times. Between two breakpoints the temperature changes linearly, which reproduces furnace ramps quoted in degrees per hour. A hold is two breakpoints with the same temperature.
- **Constant-interval kind.** A list of intervals, each with one temperature. This is the form needed for a sudden temperature step (CRP-6 case 3c, 200 h at 1600 °C and then 200 h at 1800 °C) and for the exact projection of block R3.

Keeping the two kinds apart is a DESIGN choice (decision 9), because the approved text requires strictly increasing times and a jump cannot be written as two breakpoints at one time. A schedule that mixes ramps and jumps is not needed by any benchmark in the corpus and is left out.

Table 4.4 Equations of block D.

Equation	Statement	Status	Class
(P22.21)	linear interpolation of T between breakpoints	APPROVED	DERIVED NOT IMPLEMENTED
(P22.39)	$T(t) = T_k$ on the interval $[t_k, t_{k+1})$	APPROVED	DERIVED NOT IMPLEMENTED
(P22.58)	segment length from a quoted ramp rate, or a hold time	APPROVED	DERIVED NOT IMPLEMENTED
(P22.59)	arrival and end times from a table of (temperature, time to reach, time at)	APPROVED	DERIVED NOT IMPLEMENTED

```
HistoryLinear   = { times_s : list, temps_K : list }          -- same length >= 2, times_s[1] = 0
HistoryConstant = { starts_s : list, temps_K : list, end_s }  -- starts_s[1] = 0, temps_K[k] holds on
       ↪  [starts_s[k], starts_s[k+1])
Schedule_item   = RAMP(T_target, rate)  |  HOLD(duration)     -- input of the segment builder
Triple          = { T_degC, time_to_reach_h, time_at_h }      -- CRP-6 table style
```

```
BLOCK D  temperature history                          planned module: src/thermal/temperature_history.rs
INPUT   a schedule in one of the forms above, with unit tags
OUTPUT  a History with the functions D5 to D8                                              [NEW]

D1  build_linear_from_segments(T_start, items, units)
      1. t = 0; T = T_start (converted by step A8); points = [(0, T)].
      2. FOR each item in order:
           RAMP(T_target, rate): require rate > 0 (the sign is carried by T_target - T);
                                 t = t + |T_target - T| / rate;  T = T_target;
           HOLD(duration):       require duration > 0; t = t + duration.
           append (t, T) to points.                                        (P22.58) APPROVED
         why: the length of a segment is its temperature change divided by the quoted rate.
              Rates in degC per h and K per h are equal numbers, so only the time unit needs converting.
```

```
      3. convert all times to seconds and return HistoryLinear(points).
      Example: the AGR-1 plan at 1600 degC (ASSUMPTION: start at 30 degC; the safety-test procedure
      in register rows TH-001 onward does not state the first two ramp rates; the register gives
      120 degC per h from its other source) totals 333.78 h with the plateau starting at 31.17 h, as
      printed (334 h). The 1700 and 1800 plans total 335.95 h and 338.12 h.

D2  build_linear_from_triples(triples)
      1. read "-" as 0 for both durations; take triples[1].T as the start value at t = 0.
      2. t_arr[k] = t_end[k-1] + time_to_reach[k];  t_end[k] = t_arr[k] + time_at[k];
         append (t_arr[k], T[k]) and (t_end[k], T[k]).                          (P22.59) APPROVED
         why: the table lists the time to arrive at each temperature and the time spent there.
      3. IF two consecutive points would have equal times but different temperatures THEN stop with
         "instantaneous change: use the constant-interval kind".
      4. [OPEN] The ramp between listed temperatures is taken as linear. TECDOC-1674 does not state
         the shape (NOT VERIFIED). The summed durations reproduce all nine printed totals.
      5. [OPEN] Row 8a "(1550, 6.5, -), (300, 1, -)" must be re-read from the page image before use.

D3  build_constant(starts_h, temps, end_h, units)
     convert to seconds and kelvin; require starts strictly increasing, starts[1] = 0, end > last start;
      return HistoryConstant.                                                   (P22.39) APPROVED

D4  validate(h)
      LINEAR: at least two points, all values finite, times[1] = 0, times strictly increasing, every T >
        ↪  0 K.
      CONSTANT: starts strictly increasing, every T > 0 K, end_s > last start.
      The end time of the history is times[last] (LINEAR) or end_s (CONSTANT).

D5  temperature_at(h, t)                    -- value at the START of a step or at an output time
      1. require 0 <= t <= end; never extrapolate.
      2. k = largest index with times[k] <= t (binary search).
      3. CONSTANT kind: return temps[k]  (right-continuous; at a jump the new value holds).
      4. LINEAR kind: IF t == times[k] THEN return temps[k] exactly.
         IF k is the last point THEN return temps[last].
         return temps[k] + (temps[k+1] - temps[k]) * (t - times[k]) / (times[k+1] - times[k]).
        ↪  (P22.21) APPROVED
         why: the table value itself, not a recomputed one, so that T at a breakpoint is bit exact.

D6  temperature_at_step_end(h, t_old, t_new)   -- value at the END of a step that starts at t_old
       ↪  [NEW, DESIGN]
      1. k = largest index with times[k] <= t_old.
      2. LINEAR kind: IF t_new == times[k+1] THEN return temps[k+1] exactly, ELSE return D5(h, t_new).
      3. CONSTANT kind: return temps[k]  (the value of the segment that contains the whole step).
      why: at a jump the right-continuous value of D5 at t_new belongs to the NEXT segment. A step that
           ends on the jump must be judged and computed with the old temperature. Using D5 there makes
           the step guard of block F see an impossible jump of ln D and halve the last step before every
           jump (the check script shows a shortest step of 5.6e-7 s and 66 steps instead of 36 steps).
           The LINEAR kind is continuous, so both functions agree there.

D7  slope_after(h, t)                       -- right-hand slope, used by block F
      LINEAR: k as in D5 (for t == end return 0); return (temps[k+1] - temps[k]) / (times[k+1] -
        ↪  times[k]).
      CONSTANT: return 0.
      why: at a breakpoint the slope jumps; the step that starts there lives on the new segment.

D8  events(h, other_flags)
      1. LINEAR: every breakpoint time > 0 gets the flag BREAK.
         CONSTANT: every start time > 0 gets the flag JUMP.
      2. block F adds the flags SWITCH (from block E), OUTPUT and END.
      3. require each time to lie in (0, end]; merge duplicates by exact float equality (flags are
        ↪  united); sort ascending.
         why: the event list is a set. Listing a breakpoint twice then cannot change the step sequence.
```

For the constant-interval kind, `temperature_at` returns `temps[k]` with `k` the largest index with `starts[k] <= t`, so the temperature is right-continuous and the slope is zero.

Checks already made on these steps. AGR-1 plans and ramps: totals 333.78 h, 335.95 h and 338.12 h; plateau start 31.17 h; `T(t)` reproduces every breakpoint value; the last segment ends at 30 °C. The CRP-6 conversion of the nine printed lists reproduces the printed totals (`sec5_schedules.py`, 17 of 17 PASS).

4.6 Block E: yield and source history

Origin. Decay part, block B.

Table 4.5 Equations of block E.

Equation	Statement	Status	Class
(P21.8), (P21.52)	$S_K = y\dot{F}/N_A$, kernel only	APPROVED (unit check)	DERIVED NOT IMPLEMENTED
(P21.10)	mixed yield $y = \sum_q f_q y_q$ for several fissile nuclides	APPROVED (a definition, no separate B3 row)	DERIVED NOT IMPLEMENTED
(P21.11), (P21.12)	precursor lag, $t_{99} = \ln 100/\lambda_p = 6.644\, t_{1/2,p}$	APPROVED	DERIVED NOT IMPLEMENTED
(P21.57), option A	exact step average of the source	APPROVED	DERIVED NOT IMPLEMENTED
(P21.9)	$\dot{F} = \beta n_{\mathrm{HM}}/t_{\mathrm{irr}}$ from burnup	time average APPROVED; n_{HM} NOT VERIFIED	OPEN
(P21.13) to (P21.16)	activation products	equations APPROVED; all data NOT VERIFIED	FUTURE

Independent or cumulative yield: the choice rule. The independent yield y^I counts atoms formed directly in the fission. The cumulative yield y^C also counts the atoms that arrive later by decay of the precursors. STRATO v2.0 carries one species at a time, so the precursors are not tracked. Section 3.3.2 of the derivation gives the rule, written here as a decision list.

1. If the nuclide is produced by neutron capture on a stable fission product (Ag-110m, Cs-134, Eu-154), no fission-yield source exists for it. Stop with an error that names the missing activation data. This is the FUTURE route of step E5.
2. If the precursors are not carried as separate species (always true in v2.0), use y^C. The product is then born at the moment of fission instead of after the precursor chain has decayed.
3. If a later version carries the precursor as its own species, the daughter needs y^I in its fission term. Using y^C would count the precursor twice (Section 3.3.4, Eq. (P21.18)).
4. Report the precursor delay t_{99} of Eq. (P21.12). If t_{99} is longer than a set fraction of the time to the first output, warn that the source is early: the true birth $B(t)$ stays below $y^C \dot{F} t$ by up to $1/\lambda_p$ in time.

Table 4.6 Yield treatment by product. The t_{99} values were recomputed from the half-lives by `b4_phase2_pseudocode_integration.py` (0.4228 h, 0.4761 h, 29.76 h, 9.20 d, 34.86 d); the last column is ten times t_{99}.

Product	Longest precursor	t_{99}	Treatment in v2.0	Warning if the first output is earlier than (factor 0.1)
Cs-137	Xe-137, 3.818 min	0.423 h	y^C	4.2 h
Sr-90	Rb-90m, 258 s	0.476 h	y^C	4.8 h
Kr-85	Kr-85m, 4.48 h	29.8 h	y^C with a warning in most runs	12.4 d
I-131	Te-131m, 33.25 h	9.2 d	y^C with a warning in most runs; the delay is comparable to the I-131 half-life	92 d

Product	Longest precursor	t_{99}	Treatment in v2.0	Warning if the first output is earlier than (factor 0.1)
Cs-134	Xe-133, 5.2475 d (precursor of the target Cs-133)	34.9 d	refused, activation route (FUTURE)	not applicable

The warning threshold is a proposal (decision 5, open item O2). Section 3.3.2 says only that y^C is valid when t_{99} is short compared with the time step and the diffusion time.

```
BLOCK E  yield and source history                    planned module: src/physics/fission_source.rs
INPUT   nuclide; fissile mix: list of (fissile nuclide, fraction f_q) with sum f_q = 1;
        fission rate density history: list of (start s, end s, rate in fissions per m3 per s),
          constant on each interval;
        yield table (atoms per fission) for each fissile nuclide;
        longest-precursor half-life (s) if known; warning factor (default 0.1)
OUTPUT  yield; source_kernel_i for each interval i (mol per m3 per s); the switch times; the function E3
       ↪  [NEW]

E1  Choose the yield (decision list above).
      IF the nuclide is an activation product THEN stop (see E5).
      FOR each fissile nuclide q: read y_q = cumulative yield of the product.
      Check that the fractions f_q add to 1 within 1e-9.
      yield = sum over q of f_q * y_q.                                          (P21.10) APPROVED
      why: a reactor with U-235 and Pu-239 fissions has a yield between the two pure ones.
      IF a longest-precursor half-life is known THEN
          t99 = ln(100) * precursor_half_life_s / ln(2).                        (P21.12) APPROVED
          IF t99 > warning_factor * (time of the first output) THEN print a warning with t99.   [OPEN:
        ↪  factor]
E2  FOR each interval i of the fission rate history:
        source_kernel_i = yield * fission_rate_i / N_A.                    (P21.8, P21.52) APPROVED
    Units: (atoms per fission)(fissions m^-3 s^-1)/(atoms per mol) = mol m^-3 s^-1.
    IF any rate is negative or not finite THEN stop with an error.
    why: the source enters as a right-hand-side term; a negative value voids positivity (P21.62).
    The switch times are the starts and ends of the intervals; block F receives them with the flag
       ↪  SWITCH.
E3  Function source_step(t0, t1):
        source_step = ( sum over intervals i of source_kernel_i * overlap(i, [t0, t1]) ) / (t1 - t0),
       where overlap is the length of time in common.                  (P21.57, option A) APPROVED
    why: if a switch falls inside the step, this keeps the generated amount exact; taking the value at
         the end of the step (option R) would lose or add up to |jump| * kernel volume * dt (P21.79).
         In the PLANNED mode every switch is a step end, so no switch falls inside a step; the
         function stays exact for FIXED runs and for tests.
    dt in H3 is the t_new - t_old that block F computed once; E3 is called with the same two times.
E4  (Optional, for plots) record the precursor delay t99 from E1 in the output header.
E5  FUTURE, not part of v2.0.  Activation products (P21.13 to P21.16).
      A product A is made from a stable target P by neutron capture:
          source_A(r, t) = f_m * sigma_P * flux * C_P(r, t),
      so it exists wherever target atoms are, also in the coatings.
      Needed and not yet verified: capture cross section sigma_P, isomeric fraction f_m,
      burn-out cross section of A, flux history. The time stepping would carry two species,
      target first, then product with the target's new values as source. The time-discretisation
      choice for this coupling is OPEN. The closed modal form (P21.15, P21.16) applies when A and P
      share one diffusivity per layer.
```

The burnup route. Eq. (P21.9) turns burnup β (FIMA) into a mean fission rate density only with the heavy-metal atom density n_{HM} of the kernel, which no audited source gives. A conversion from reactor power would need the energy released per fission, which the sections do not contain. Phase 2 therefore takes the fission rate density as a direct input (decision 2). Both conversions stay OPEN.

Synthetic example. For Cs-137 with $y^C = 6.2208 \times 10^{-2}$ (JEFF-3.1.1, U-235, thermal; register Table 22) and a synthetic $\dot{F} = 5.0 \times 10^{19}$ fissions m^{-3} s^{-1}, step E2 gives $S_K = 5.1649 \times 10^{-6}$ mol m^{-3} s^{-1} (recomputed from `fission_yields.csv` by the integration script). The rate density is a test value and not a STRATO input.

4.7 Block F: events and step planner

Origin. This block merges two drafts that each had their own step planner: block C1 of the decay part (a list of steps built from the decay and diffusion limits, growing geometrically after each boundary) and block T3 of the temperature part (steps chosen one at a time from the slope of $T(t)$, landing on each event). The two disagreed on the growth factor (1.5 against 2.0), on how the last step before an event is shaped (a common scale factor against halving the remainder), on whether a step restarts small after an event, and neither reproduced the v1.0 constant grid. The merged rules are DESIGN choices that keep the rules of both parts that the equations support (decision 10; mismatches X1, X2 and X7 in Section 8).

Table 4.7 Equations of block F.

Equation	Statement	Status	Class
(P21.64), (P21.77)	Backward Euler decay error $e^{-z} - 1/(1+z) = -z^2/2$; step rule $\Delta t \le 2\varepsilon_\lambda/(\lambda^2 t_{\text{end}})$	APPROVED	DERIVED NOT IMPLE-MENTED
(P21.78)	slowest-mode error $\frac{1}{2}(\Lambda_1 + \lambda)^2 t \Delta t$	APPROVED for constant diffusivities; OPEN for a temperature history	DERIVED NOT IMPLE-MENTED
(P21.79) and its rules	amount error at a switch inside a step; place a step end on every switch, output time and temperature knot; restart small after a switch; no stability limit	APPROVED	DERIVED NOT IMPLE-MENTED
(P22.19), (P22.20)	$d \ln D/dt = Q_{\text{eff}}\dot{T}/(R_g T^2)$; step bound $\varepsilon_D R_g T^2/(Q_{\text{eff}}\lvert\dot{T}\rvert)$	APPROVED	DERIVED NOT IMPLE-MENTED
(P22.21)	linear $T(t)$ between breakpoints	APPROVED	DERIVED NOT IMPLE-MENTED

The temperature step bound is a resolution rule for the coefficients. Accuracy of the solution is checked separately, by comparing a run with steps Δt and $\Delta t/2$. The guard of step F5.5 compares actual logarithms of the diffusivity:

$$\max_i \left| \ln D_i(T_{\text{end}}) - \ln D_i(T_{\text{old}}) \right| \le 2\varepsilon_D \qquad \text{(H22.1)}$$

Here T_{old} is the temperature at the start of the step and T_{end} the temperature at its end (step D6). Eq. (H22.1) is a DESIGN relation: (P22.20) uses the activation energy at the start of the step, a strongly curved law can change faster than predicted, and the factor 2 leaves room for that first-order estimate.

```
BLOCK F  events and step planner                    planned modules: finite_volume.rs,
       ↪  temperature_history.rs
INPUT   t_end; history (block D) or none; switch times (block E); output times or output_every_steps;
        grid mode FIXED or PLANNED;
        FIXED: dt_request;
        PLANNED: dt_max; dt_first; growth (default 2.0); eps_D (default 0.1);
                 eps_lambda (default 1e-3); optional Lambda_1 (1/s) with eps_1; step_limit
OUTPUT  F1 to F4 once: the event list and the fixed step caps
        F5 for each step: t_old, t_new, dt, flags                                  [NEW]

F0  IF the grid mode is FIXED THEN
        n_steps = ceil(t_end / dt_request) ; dt = t_end / n_steps ;
        step k runs from (k-1)*dt to k*dt ; rows at every output_every_steps steps and at the last step.
       ↪  [KEPT]
        Leave the block; steps F1 to F5 do not run.
```

```
    why: this is the v1.0 rule (lines 190, 191 and 221). Building the same grid by the geometric rule
         with growth 1 gives the same numbers only to round-off, and the regression cases V1 and VT16
         demand bit identity. Block A has already refused FIXED when an event time exists.
F1  Events.  Collect, each with its flags:
      BREAK or JUMP   from block D, step D8,
      SWITCH          every start and end of the source history (block E), where the rate changes,
      OUTPUT          every requested output time,
      END             t_end.
    Require every time in (0, t_end]. Merge equal times by exact float equality (union of the flags).
        ↪  Sort.
    why: a step that straddles a switch smears it over one step, and a step that straddles a kink of
         T(t) loses the smooth behaviour on which the error estimate rests. Ending steps on events
         costs nothing (rule 1 under P21.79; MOOSE does the same with force_step_every_function_point).
         The list is a set, so listing an event twice cannot change the step sequence.
F2  IF decay_constant > 0 THEN
        dt_decay = 2 * eps_lambda / (decay_constant^2 * t_end).                  (P21.77) APPROVED
    ELSE dt_decay = infinity.
    why: the Backward Euler decay factor 1/(1+z) is off from exp(-z) by z^2/2 per step, so the relative
         decay error after t_end is about decay_constant^2 * t_end * dt / 2. The limit does not apply
         at a steady state, where the fixed point is exact.
F3  IF Lambda_1 is given THEN
        dt_diff = 2 * eps_1 / ((Lambda_1 + decay_constant)^2 * t_end).           (P21.78, solved for dt)
    ELSE dt_diff = infinity.
    [Temperature history, note updated 2026-11-13 from Section 5.2.5 of the derivation; the step itself
        ↪  is
     unchanged. Use Lambda_1 at the highest temperature. With a shared scaling factor and option A this
     dt_diff is then a sufficient cap: the slow-mode error stays below eps_1 (PROVED, the F3 paragraph
     under (P22.72)), at about [(Lambda_max + decay_constant) * t_end / Theta_end]^2 times the fewest
     steps (1.11 in the ramp-and-hold test). For option E the lag L_1 enters ((P22.65), (P22.73)), so
     eps_1 is not proved for this value. Lambda_1 at the lowest temperature gives no bound. With
    layer-specific activation energies only the frozen-coefficient part of the error is bounded (P22.78;
     option A for any history, option E on heating only) and the remainder is OPEN, so there the value
     stays a warning. The cheaper rule (P22.69) with the step (P22.72) is proposed in
     Section 9.5 and is not applied here. Hard cap or warning: decision 6.]
    why: the slowest diffusion mode carries the largest time error; quantities near complete release
         (for example 1 - F) are the most sensitive.
F4  dt_cap = smallest of dt_max, dt_decay, dt_diff.
    IF t_end / dt_cap is larger than step_limit THEN stop and report which of the three limits controls
        ↪  it.
    why: Backward Euler has no stability limit; these are accuracy limits only (decision 6 asks
         whether the limits of F2 and F3 are hard caps or warnings).
    Set grow_cap = dt_first and event_pointer = 1.

F5  next_step(t_old, grow_cap, event_pointer)
      1. t_event = events[event_pointer].time          -- the next event strictly after t_old
      2. s = slope_after(history, t_old)                (0 if no history)               (block D, D7)
      3. IF s == 0 THEN dt_T = infinity
         ELSE T = temperature_at(history, t_old)
              Qmax = maximum over all layers i of apparent_activation_energy(law_i, T)          (block
        ↪  C, C4)
             dt_T = eps_D * Rg * T^2 / (Qmax * |s|).                          (P22.19, P22.20) APPROVED
           why: d(ln D)/dt = Q_eff * dT/dt / (Rg T^2); fixing the allowed change of ln D fixes the step.
                A hold has no limit from this rule, a steep cool-down has a short one.
                Example values (eps_D = 0.1; Cs in SiC): 1600 degC at 50 K/h gives 479 s;
                1250 degC at 120 K/h gives 445 s; 1600 degC at -600 K/h gives 40 s.
      4. dt = minimum of dt_cap, grow_cap, dt_T.
         why: the growth cap keeps the step from jumping by orders of magnitude when a slope changes.
      5. LOOP (at most 30 times):
           rem = t_event - t_old
           IF rem <= dt THEN t_new = t_event                    (this step lands on the event, by
        ↪  assignment)
           ELSE IF rem <= 2*dt THEN t_new = t_old + rem / 2     (halve the remainder)
           ELSE t_new = t_old + dt
           why: the middle case replaces "one full step plus a sliver" by two equal steps, so no step
                is much shorter than its neighbours and no step overshoots the event.
           Guard (H22.1): T_old = temperature_at(history, t_old);
                          T_end = temperature_at_step_end(history, t_old, t_new)          (block D, D6)
                           compare the layer logarithms of the diffusivity; skip the guard if every law is
        ↪  constant.
           IF the guard holds OR t_new - t_old < 1 microsecond THEN leave the loop
           ELSE dt = (t_new - t_old) / 2 and repeat; after 30 repeats stop with an error.
      6. dt_n = t_new - t_old, computed once by this single subtraction.
         why: the very same dt_n must enter the matrix, the source average and the release bookkeeping,
              otherwise the balance does not close.
      7. IF t_new == t_event (it was assigned, never computed by addition) THEN
             event_pointer = event_pointer + 1 ; flags = events[...].flags
```

```
            grow_cap = dt_first IF the flags contain SWITCH or JUMP, ELSE growth * dt_n
          ELSE flags = none ; grow_cap = growth * dt_n.
          why: after a switch or a jump the time derivatives jump (rule 2 under P21.79: restart small and
               grow geometrically); Backward Euler is L-stable, so no oscillation follows. A kink of a
               linear history (BREAK) and an output time do not restart the step.
      8. RETURN t_new, dt_n, flags, grow_cap, event_pointer.
```

Examples (calculated). With $\varepsilon_\lambda = 10^{-3}$ and 620 effective full-power days, Ag-110m ($\lambda = 3.211198 \times 10^{-8}$ s^{-1}) gives $\Delta t \leq 3.62 \times 10^4$ s, about 10 h. I-131 over five mean lives gives $\Delta t \leq 400$ s. For Cs-137 the limit is far beyond any heating test. These agree with Section 4.6 of the derivation.

Checks already made on these steps (Python). On the AGR-1 1600 °C plan (41 events; layers Cs in UO_2, constant buffer, PyC, Cs in SiC, PyC; $\varepsilon_D = 0.1$, $\Delta t_{\max} = 3600$ s, growth 2) the temperature-part rule gave 1826 steps, every event a step end by exact float equality, a largest change of $\ln D$ of 0.102, no halving by the guard, a shortest step of 1.24 s, and a step sequence that does not change when a breakpoint is listed twice (`b4_arrhenius_checks.py`). The merged rule was run by `b4_phase2_pseudocode_integration.py` on a ramp from 1000 to 1800 °C in 4 h with a 4 h hold, a source switched off at 3 h and $\lambda = 10^{-5}$ s^{-1}: 139 steps, all 8 events are step ends by float equality, the step after the switch restarts at $\Delta t_{\text{first}} = 5$ s, the largest change of $\ln D$ is 0.102 and the growth between steps never exceeds 2.

4.8 Block G: geometry once

Origin. Temperature part T4.1, and step B3 of the decay part (the cell factor).

Table 4.8 Equations of block G.

Equation	Statement	Status	Class
(P21.53), (P21.54)	cell average of the source over the cell volume; for the kernel source: factor 1 inside the kernel, 0 outside, $(R_1^3 - a^3)/(b^3 - a^3)$ for a cell cut by R_1	APPROVED	DERIVED NOT IMPLEMENTED
(P22.14)	geometry factors A_f, $\delta_f^{\pm}$ of the conductance	APPROVED	DERIVED NOT IMPLEMENTED

```
BLOCK G  geometry                                        planned module: src/numerical/finite_volume.rs
       ↪  (build_mesh split)
INPUT   regions: outer radius, cells; kernel radius R1 = outer radius of region 1
OUTPUT  per cell: volume, centroid, layer index, cell_factor; per face: area, dist_left, dist_right
       ↪  [CHANGED]

G1  FOR each cell j (edges left, right; layer index = region):                                  [KEPT]
        volume[j]   = (4 pi / 3) * (right^3 - left^3)
        centroid[j] = 0.75 * right                                      IF left == 0
                      0.75 * (right^4 - left^4) / (right^3 - left^3)    otherwise
G2  FOR each internal face f between cell j and j+1 (radius r_f = right edge of cell j):
        area[f]       = 4 pi r_f^2
        dist_left[f]  = r_f - centroid[j]
        dist_right[f] = centroid[j+1] - r_f
G3  Outer face: area_R = 4 pi R^2 ; dist_R = R - centroid[N]
G4  FOR each cell j: cell_factor[j] =
        1                                       IF right <= R1
        0                                       IF left >= R1
        (R1^3 - left^3) / (right^3 - left^3)    otherwise.                          (P21.54) APPROVED
    On the v1.0 mesh every layer boundary is a cell face, so the third case never occurs.
    IF it occurs anyway THEN keep the formula (it is exact) and print a note.
    why: the cell average, not the value at the cell centre, is what the balance needs. For the v1.0 mesh
         the sum of volume * cell_factor equals (4 pi / 3) R1^3 exactly (P21.54).
G5  Nothing here depends on D, lambda or T, so nothing here is recomputed inside the loop.
    why: v1.0 build_mesh fixes the conductances while it builds the mesh (lines 78 to 91); only the
```

```
        part that does not depend on D is kept here, and block H2 does the rest.
```

4.9 Block H: one time step

This is the heart of the loop. Four steps prepare the numbers; the fourth solves the system.

4.9.1 What changes with respect to v1.0 The v1.0 function `backward_euler_step` (lines 162 to 182) builds a tridiagonal system from three arrays of face conductances and one array `v`. Two lines change; the others stay.

Table 4.9 What changes in one Backward Euler step.

Item	v1.0.0 (lines 178 and 179)	Phase 2
diagonal excess `v[i]`	`volume / dt`	`volume / dt + lambda * volume`
right-hand side `rhs[i]`	`v[i] * old[i]`	`(volume / dt) * old[i] + volume * source_cell[i]`
face conductances	`conductance`, `outer_conductance` of the mesh	recomputed in H2 for each step
solver (subtraction-free elimination)	lines 132 to 160	unchanged
layout of the full matrix row	$-g_j^W C_{j-1} + (v_j + g_j^W + g_j^E) C_j - g_j^E C_{j+1}$	same form with the new v_j

Two traps follow from Section 4.3 of the derivation.

- After `v` is redefined, the right-hand side must still use $V_j/\Delta t$. Reusing the enlarged `v[i]` as in line 179 would multiply the old inventory by $1 + z$ and the run would lose inventory without any decay having been asked for.
- Write the diagonal as `volume / dt + lambda * volume`. With $\lambda = 0$ the second term is exactly 0.0 and the result is bit-identical to v1.0. The algebraically equal `volume * (1/dt + lambda)` differs from `volume / dt` in the last bit for 13 of the 40 cells of the test mesh and would break the regression test V1. Rust does not fuse a multiply and an add unless the program asks for it (`mul_add`); the Rust test V1 is what confirms this for the real compiler.

Table 4.10 Equations of block H.

Equation	Statement	Status	Class
(P21.55)	semi-discrete cell balance with decay and source	APPROVED	DERIVED NOT IMPLEMENTED
(P21.58), (P21.59)	Backward Euler system; diagonal $V_j(1/\Delta t + \lambda)$, right-hand side $(V_j/\Delta t)C_j^n + V_j \bar{S}_j^{n+1}$	APPROVED	DERIVED NOT IMPLEMENTED
(P21.60), (P21.61)	diagonal excess and subtraction-free pivots stay valid	APPROVED	v1.0 solver IMPLEMENTED, lines 132 to 160
(P21.62), (P22.17)	positivity for every $\Delta t > 0$, $\lambda \geq 0$ and every temperature	APPROVED	DERIVED NOT IMPLEMENTED
(P21.63)	link to the v1.0 step: $(1+z)^{-1}$ times the v1.0 step with $\Delta t' = \Delta t/(1+z)$	APPROVED	DERIVED NOT IMPLEMENTED
(P21.80)	decay factor $c_\lambda = e^{-\lambda t} c_0$, valid also for a diffusivity that changes in time	APPROVED	DERIVED NOT IMPLEMENTED
(P22.13) to (P22.15)	semi-discrete and Backward Euler form with time-dependent conductances	APPROVED	DERIVED NOT IMPLEMENTED

Equation	Statement	Status	Class
(P22.18)	difference between the end-of-step and step-average choices	APPROVED	DERIVED NOT IMPLEMENTED
(P22.22), (P22.23)	integral of an Arrhenius term over a linear ramp, exponential integral E_2	APPROVED	DERIVED NOT IMPLEMENTED

Decay and a changing diffusivity. Decay enters only the diagonal term λV_j of Eq. (P21.59), so block H needs no change when the conductances are rebuilt in every step. The reason is Eq. (P21.80): for a source-free problem the solution with decay is $e^{-\lambda t}$ times the solution without decay, and the proof of that statement uses only the linearity of the spatial operator, so it holds when $D_i = D_i(t)$. This is also why block R2 can split off the factor $e^{-\lambda t}$ before the clock change. The identity needs one decay constant for the whole particle, constant in time (the same restriction as condition E3 of block R2).

4.9.2 Which diffusivity does a step use Two ways exist to pick the diffusivity of a step.

- **Option E, end of step:** D_i evaluated at the temperature at the end of the step (step D6). This is the textbook fully implicit choice and the one named in the task brief.
- **Option A, step average:** $\bar{D}_i = \frac{1}{\Delta t_n} \int_{t_n}^{t_{n+1}} D_i(T(t))\, dt$. The approved text (Section 5.2.3) recommends this option, because for a shared scaling factor and no decay it is exactly Backward Euler in the effective time, and because it stays exact across a kink of $T(t)$.

Both are first order in time. Neither is uniformly more accurate in the tests (A is 2.24 times better in the shared-α ramp, E is 1.75 times better in the layer-specific test). The default here is A and E stays selectable (decision 8). In the joined check run (ramp from 1000 to 1800 °C, decay, switched source) the released amounts differ by 1.4 % between the two options at the planner's step sizes, which shows that the choice matters at these steps. That run has no reference solution, so it does not say which option is closer to the truth.

```
BLOCK H  one time step                                  planned module: src/numerical/finite_volume.rs
       ↪  (extended), diffusivity.rs
INPUT   t_old, t_new, dt (from block F); conc[j] (old); decay_constant; laws; history; option A or E;
        geometry and cell_factor (block G); source_history (block E)
OUTPUT  conc_new[j]; and for block J: G_R, gen_rate                                          [CHANGED]

H1  step_diffusivity(t_old, t_new)  ->  D_layer[1..5]
      1. IF no history is given THEN D_layer[i] = diffusivity(law_i, T_const)  and leave the step.
      2. require [t_old, t_new] to lie inside one history segment (block F guarantees it).
      3. Option E: T_end = temperature_at_step_end(history, t_old, t_new)                (block D, D6)
                   D_layer[i] = diffusivity(law_i, T_end).
         why: the left-limit value; the right-continuous value belongs to the next segment.
      4. Option A:
           IF the segment slope is 0 THEN D_layer[i] = diffusivity(law_i, temperature_at(history,
       ↪  t_old)); no quadrature.
           ELSE FOR each layer i and each branch m:
                arg_branch = (Q[m] / Rg) * | 1/T_end - 1/T_old |
                IF arg_branch <= 1 THEN Dbar_m = 8-point Gauss-Legendre average of the branch over
       ↪  [t_old, t_new]
                ELSE Dbar_m = (1/(t_new - t_old)) * D0_eff * (1/s) * [ T * E2(theta/T) ] from T_old to
       ↪  T_end,
                              with theta = Q[m]/Rg, s = the slope                            (P22.22)
       ↪  APPROVED
                D_layer[i] = sum over m of Dbar_m ; then apply the floor of step C3.6.
         why: the average of a sum is the sum of the averages. Eight-point Gauss-Legendre has a relative
              error below 1e-15 while arg_branch <= 1 (checked: 4.8e-16 at a change of 2 in ln D, 1.9e-13
```

```
            at 4, 5e-9 at 8). The closed form keeps full accuracy for large changes (2e-15 or better from
              0.5 upward; 2.3e-15 at a change of 16 in the integration check) but loses digits on pieces
              below about 1e-3 K (relative error 2.6e-11 at 1e-3 K, 8e-9 at 1e-5 K). Steps from block F
              change ln D by at most 2 * eps_D, so Gauss-Legendre is the normal path.
          E2(x) = exp(-x) - x * E1(x), E1 the exponential integral of order 1.                (P22.23)
        ↪  APPROVED
       5. RETURN D_layer.

H2  conductances(D_layer)       -- O(N) per step
      FOR each internal face f between layer a (left) and layer b (right):
          g[f] = area[f] / ( dist_left[f] / D_layer[a] + dist_right[f] / D_layer[b] )        (P22.14)
        ↪  APPROVED
      G_R   = area_R * D_layer[5] / dist_R                                    (P22.14) APPROVED
      west_g[j] = 0 for j = 1, g[j-1] otherwise ;  east_g[j] = g[j] for j < N, G_R for j = N.
      why: two half-cell resistances in series, as in v1.0. A face inside one layer uses the same D on
        ↪  both sides.
           A constant temperature needs this once; the planned code may reuse the arrays while D_layer
        ↪  is unchanged.

H3  source_for_step(t_old, t_new, dt)  ->  source_cell[j], gen_rate
      1. source_step = block E, function E3, called with t_old and t_new.                   (P21.57,
        ↪  option A)
      2. source_cell[j] = source_step * cell_factor[j].
      3. gen_rate = sum over j of volume[j] * source_cell[j].
      why: for the v1.0 mesh gen_rate equals (4 pi / 3) R1^3 * source_step exactly (P21.54).
           IF there is no source, source_cell[j] = 0 for every j.

H4  one Backward Euler step with decay and source
      1. Check: dt finite and > 0; decay_constant finite and >= 0; every array has N entries;
         every conc_old[j] and source_cell[j] >= 0.
         (A negative source is allowed only in a manufactured-solution test, with an explicit switch.)
         why: positivity (P21.62) needs a nonnegative old state and source.
      2. FOR each cell j:
             diag_excess[j] = volume[j] / dt + decay_constant * volume[j]        [CHANGED] (P21.59)
             rhs[j]         = (volume[j] / dt) * conc_old[j] + volume[j] * source_cell[j]   [CHANGED]
        ↪  (P21.59)
           why: the first term of rhs is the old inventory of the cell divided by dt, the second is the
        ↪  atoms
                born in the cell per second; decay acts like an extra conductance from every cell to a
                reservoir that stays at zero concentration, so it only enlarges the diagonal excess
        ↪  (P21.60).
      3. Run the v1.0 elimination on (diag_excess, west_g, east_g, rhs).             [KEPT] (P21.61)
         Forward sweep, for j = 1..N:
             pivot_e[j] = diag_excess[1]                          IF j = 1
            pivot_e[j] = diag_excess[j] + west_g[j]*pivot_e[j-1] / (pivot_e[j-1] + west_g[j])   otherwise
             den[j]     = pivot_e[j] + east_g[j]
             carry[j]   = west_g[j] * partial[j-1]              (0 for j = 1)
             partial[j] = (rhs[j] + carry[j]) / den[j]
         Back substitution: conc_new[N] = partial[N];
             conc_new[j] = partial[j] + east_g[j] / den[j] * conc_new[j+1], for j = N-1 down to 1.
         why: this form contains only sums, products and quotients of nonnegative numbers. The textbook
              pivot subtracts two nearly equal numbers when diffusion is fast. Decay only makes every
        ↪  pivot
             larger, so the proof of v1.0 holds (P21.61). With conductances that change from step to step
              the matrix is still an M-matrix (strictly dominant diagonal, non-positive off-diagonals).
        ↪  (P22.17)
      4. RETURN conc_new.
        Property: conc_new[j] >= 0 for every dt > 0, decay_constant >= 0 and every temperature, in exact
         arithmetic and in floating point, because no step subtracts (P21.62, P22.17).
```

Worked scalar example (calculated). Take one cell without diffusion, $C^n = 1$, $S = 10^{-7}$ mol m^{-3} s^{-1}, $\lambda = 10^{-6}$ s^{-1} and $\Delta t = 10^5$ s, so $z = 0.1$. Step H4 gives $C^{n+1} = (C^n + \Delta t S)/(1+z) = 0.9181818$. The exact solution of $dC/dt = -\lambda C + S$ is $C^n e^{-z} + (S/\lambda)(1 - e^{-z}) = 0.9143537$. The difference of $+3.8 \times 10^{-3}$ is the local error of this single step. It is of order z^2, as Eq. (P21.64) predicts for the decay factor (to leading order it is $(C^n - S/\lambda)z^2/2$, here 4.5×10^{-3}, with the higher terms making up the rest), so halving Δt divides the one-step difference by about 4 (3.7 in the B7 check). Over a fixed time interval the number of steps doubles as well, so the accumulated error falls in proportion to Δt: the scheme is first order.

Why the unsplit step is kept. The alternative of taking a v1.0 step and then multiplying by e^{-z} (Eq. (P21.66)) is exact in the decay but its steady state is shifted by the factor $\varphi(z) = ze^z/(e^z - 1)$ in the diffusion term (Eq. (P21.68)). Its steady release-to-birth ratio (balance reading) is wrong by up to +127 % in the test cases of Section 4.4 of the derivation, depending on λ/Λ_1 and z. The unsplit step keeps the exact semi-discrete steady state for any Δt (checked: a step of any size from 10^3 to 10^{12} s changes the steady profile by less than 10^{-15} relative). The split form stays in the plan as a cross-check for $S = 0$ (case V15).

Relation to the effective time. For a shared scaling factor and $\lambda = 0$, option A equals Backward Euler in the effective time with step $\Delta\tau_n = \int_{t_n}^{t_{n+1}} \alpha\, dt$, because every conductance is then $\Delta\tau_n/\Delta t_n$ times its constant-temperature value. The integration check finds a relative difference of 4×10^{-16} in the inventory after a ramp and hold with five layers. For $\lambda > 0$ the decay stays inside the implicit operator and the two differ at first order (2.7×10^{-3} after 30 steps in the B3 check), so no test may demand equality there.

4.10 Block J: bookkeeping

Origin. Decay part, block D, and step T4.5 of the temperature part.

Release is counted two ways. M_{rel} is every atom that has ever crossed the surface. M_{out} is the part that is still undecayed outside. They agree only for $\lambda = 0$ (Eq. (P21.76)). A comparison with a benchmark number has to say which one the benchmark means. TECDOC-1674 Eq. (7.8) is $F_B = M_{\text{out}}/(M + M_{\text{out}})$ (Section 3.7.3 of the derivation), and the decay-corrected heating-test fraction is $F_{\text{corr}} = e^{\lambda t} M_{\text{out}}/M_0$.

Table 4.11 Equations of block J.

Equation	Statement	Status	Class
(P21.40) to (P21.42)	$dM/dt = \text{Gen} - \lambda M - J_R$; closure $M_0 + M_{\text{gen}} = M + M_{\text{dec}} + M_{\text{rel}}$	APPROVED	DERIVED NOT IMPLEMENTED
(P21.43), (P21.44)	atoms outside the particle: $M'_{\text{out}} = J_R - \lambda M_{\text{out}}$	APPROVED	DERIVED NOT IMPLEMENTED
(P21.45)	release fractions F_{rel}, F_{corr}, F_{gen}, R/B, F_B, F_{post}	APPROVED (definitions)	DERIVED NOT IMPLEMENTED
(P21.70), (P21.71)	per-step identity $M^{n+1} - M^n = \Delta t(\text{Gen}^{n+1} - \lambda M^{n+1} - J_R^{n+1})$	APPROVED	DERIVED NOT IMPLEMENTED
(P21.72)	summed closure with M_{gen}, M_{dec}, M_{rel}	APPROVED	DERIVED NOT IMPLEMENTED; v1.0 residual IMPLEMENTED (line 220)
(P21.73), (P21.75), (P21.76)	Backward Euler update of M_{out}; closure with $M_{\text{dec,out}}$	APPROVED	DERIVED NOT IMPLEMENTED
(P21.74)	exponential update of M_{out}	APPROVED	optional (decision 4)
(P22.16)	summed step identity with the G_R of the step	APPROVED	DERIVED NOT IMPLEMENTED

v1.0 stores the inventory, one cumulative release sum with Kahan compensation (lines 211 to 217) and the residual $M + M_{\text{rel}} - M_0$. It rejects a run whose initial inventory is not positive (lines 186 to 189); that check has to go, because irradiation from zero is

a case with $M_0 = 0$, and `release_fraction = released / initial` has to be guarded against a zero denominator.

```
BLOCK J  bookkeeping                                  planned module: src/physics/inventory.rs
INPUT   conc_new[j], G_R, gen_rate, dt (all from the same step); volumes; decay_constant
OUTPUT  running totals and residuals                                                  [CHANGED]

J0  Once, before the loop:
      inv0 = sum over j of volume[j] * conc[j].                                       [KEPT]
      gen_total = dec_total = rel_total = dec_out_total = 0   (each a Kahan compensated sum);
      out_now = 0.
      IF inv0 = 0 AND the source is zero everywhere THEN stop: there is nothing to follow.
      why: v1.0 stops whenever inv0 is not positive; v2.0 only stops when nothing is in the particle
           and nothing will be made.
J1  For each step:
      1. inv = sum over j of volume[j] * conc_new[j].                                 [KEPT]
      2. release_rate = G_R * conc_new[N], with the G_R that block H2 used in this step.   (P21.71,
        ↪  P22.16) [CHANGED]
        why: v1.0 line 210 reads the fixed mesh.outer_conductance. That is correct for a constant D only.
             With a changing D the release must use the G_R of the step, or the balance fails (the
             negative test in Section 7 gives a residual of 1.2e-5).
      3. Add to the compensated sums (Kahan, as v1.0 does for the release):
            gen_total += dt * gen_rate                                  (P21.72)  [NEW]
            dec_total += dt * decay_constant * inv                      (P21.72)  [NEW]
            rel_total += dt * release_rate                              (P21.72, P22.16)  [KEPT]
         why: each increment is the term that appears in the per-step identity (P21.70), so the summed
              identity (P21.72) closes to round-off.
      4. Atoms outside the particle:
            out_now = (out_now + dt * release_rate) / (1 + decay_constant * dt)      (P21.73) [NEW]
            dec_out_total += dt * decay_constant * out_now                           (P21.75) [NEW]
        why: the outside inventory obeys M_out' = J_R - lambda M_out; this is the Backward Euler form of
             that law, the same rule that is used inside, so the closure below holds with intrinsically
             defined terms.
         Optional report value (only if outside decay needs better than first order; decision 4):
            out_exp = exp(-decay_constant*dt) * out_exp_old
                      + release_rate * dt * phi1(decay_constant*dt)                  (P21.74, H21.1)
         (phi1 is defined in Eq. (H21.1) of Section 5.1.) In that case dec_out_total must be taken as
         the remainder out_old + dt*release_rate - out_new, otherwise the closure does not hold.
      5. Residuals, relative to the total supply (inv0 + gen_total):
            res_inside  = (inv0 + gen_total) - (inv + dec_total + rel_total)                  (P21.72)
            res_outside = (inv0 + gen_total) - (inv + out_now + dec_total + dec_out_total)    (P21.75)
         IF the relative size of either exceeds 1e-12 THEN report a closure failure.   [CHANGED]
         why: the criterion is that of v1.0. The residual tests assembly and solver, and stays at
              round-off for any dt, so it does not measure time accuracy.
      6. Positivity: minimum of conc_new[j] over j. IF it is below 0.0 THEN report a failure.
        ↪  (P21.62, P22.17)
         why: the proofs say the minimum is at least 0.0 exactly; the v1.0 unit test accepts -1e-13
        ↪  (line 310), which is looser than needed.
      7. IF t_new is an output time THEN store inv, the totals, release_rate, gen_rate and the residuals
        ↪  for block K.
J2  Fractions for the output rows (P21.45), with t the output time:
      source-free heating:        F_rel  = rel_total / inv0
                                  F_corr = exp(decay_constant * t) * out_now / inv0
                                  (skip F_corr when decay_constant * t > 700, where the exponential
        ↪  overflows)
      irradiation from zero:      F_gen  = rel_total / gen_total
                                  R_over_B = release_rate / gen_rate
                                  F_B    = out_now / (inv + out_now)
      heating after irradiation:  F_post = (rel_total - rel_total_at_t_irr) / inv_at_t_irr
                                  (store rel_total and inv at the step that ends on t_irr)
      An entry is left empty when its denominator is 0.
    why: decay-corrected and uncorrected fractions answer different questions (Section 3.6 of the
         derivation). F_B equals M_out/(M + M_out) and is not M_rel/M_gen.
```

Remarks on block J.

- With a Backward Euler update the decay-corrected fraction F_{corr} is slightly off the decay-free fraction and can exceed 1 by a first-order amount. In the check it is 1.000796 on the first mesh and 1.000398 on the next for a decay-free value close to 1. This excess is the Backward Euler decay error of Eq. (P21.77); the sign of the scheme is correct.
- The residuals of step J1.5 are insensitive to the step size. A good closure is necessary and says nothing about accuracy in time. Accuracy comes from the com-

parisons of Section 7.

- When the source is switched off at t_{irr}, the profile at the end of the step that ends on t_{irr} is the starting profile of the heating phase (Eq. (P21.23)). Nothing is re-projected; the run continues with `source_cell = 0`.
- Summing the step equations over j cancels every interior face and leaves exactly the three terms of step J1.3 (Eq. (P22.16)). That is why G_R, dt and $\bar{S}$ must be the ones used in the matrix.

4.11 Block K: output rows and the end-of-run report

```
BLOCK K  output                                  planned modules: src/physics/inventory.rs, src/main.rs
INPUT   stored values from block J
OUTPUT  the output table and the report                                                    [CHANGED]

K1  At an output time write one row with: t, T (if a history exists), inv, gen_total, dec_total,
    rel_total, out_now, dec_out_total, release_rate, gen_rate, res_inside, res_outside,
    minimum of conc, and the fractions of J2.  Leave an entry empty when its denominator is 0.
    why: v1.0 writes six columns (inventory, cumulative release, release fraction, release rate,
         balance residual, minimum concentration); the new columns are the ones the new bookkeeping
       ↪  needs.
K2  End-of-run report:
      - largest |res_inside| and |res_outside| over the run and the step where they occurred;
      - smallest concentration over the run;
      - number of steps, the cap that controlled the step size (dt_max, dt_decay, dt_diff or dt_T),
        shortest and longest step;
      - number of D-floor events (step C3.6) and every window warning (step C3.5);
      - the precursor warning of step E1, if any;
      - for an effective-time or modal reference: the covered eigenvalue and the first valid time.
    why: a run that closes but was cut off by a floor or a window must say so.
```

5 Reference solutions (outside the finite-volume loop)

These blocks give numbers that the finite-volume run is compared with. None of them runs inside the loop of block M.

5.1 Block R1: modal solution with decay and a piecewise-constant source

Origin. Decay part, block E.

Scope. R1 gives the semi-analytical series for the case that v1.0 already treats: a source and an initial inventory that are uniform in the kernel, constant diffusivities, an absorbing surface. It needs only the eigenvalues Λ_n and the weights w_n that the v1.0 `ModalRelease` already computes (src/analytical/sturm_liouville.rs, lines 308 to 395; weight formula at line 352). For a general source or initial profile the block also needs the norms N_n and the moments I_n, $I_n^{(1)}$; that is the FUTURE extension of step R1.1.

Table 5.1 Equations of block R1.

Equation	Statement	Status	Class
(P21.24) to (P21.26)	eigenproblem and orthogonality; decay does not change it	APPROVED	v1.0 eigenvalues IMPLEMENTED
(P21.28) to (P21.30)	modal ODE $da_n/dt = -(\Lambda_n + \lambda)a_n + s_n$; kernel-uniform source	APPROVED	DERIVED NOT IMPLEMENTED
(P21.31) to (P21.34)	solution for any, constant and piecewise-constant source	APPROVED	DERIVED NOT IMPLEMENTED
(P21.35), (P21.37), (P21.38)	inventory, direct flux series, direct release series	APPROVED	DERIVED NOT IMPLEMENTED

Equation	Statement	Status	Class
(P21.39)	balance form $M_{\text{rel}} = M_0 + M_{\text{gen}} - M - M_{\text{dec}}$	APPROVED	DERIVED NOT IMPLEMENTED
(P21.44)	$M+M_{\text{out}} = M_0 e^{-\lambda t} + \int G e^{-\lambda(t-s)} ds$	APPROVED	DERIVED NOT IMPLEMENTED
(P21.30), (P21.46)	$4\pi a_n(0) I_n = M_0 w_n$ and, for the kernel-uniform source, $4\pi s_n I_n = \text{Gen}\, w_n$	APPROVED	v1.0 weights IMPLEMENTED
(P21.48), (P21.49)	single-sphere Booth forms used as reference	APPROVED	reference only
(H21.1) to (H21.3)	stable evaluation of the interval factors	HELPER, proved in the check script	helper

Helper identities. Let $\kappa_n = \Lambda_n + \lambda$ (`mode_kappa`), $x = \kappa_n \Delta$ (`arg_mode`), and define the two interval factors

$$\varphi_1(x) = \frac{1 - e^{-x}}{x}, \qquad \varphi_2(x) = \frac{x - 1 + e^{-x}}{x^2} \tag{H21.1}$$

For the modal inventory $m_n = 4\pi I_n a_n$ with the kernel source Gen constant on an interval of length Δ, Eqs. (P21.33) and (P21.38) can be written

$$\begin{aligned} m_n(t_j) &= m_n(t_{j-1}) e^{-\kappa_n \Delta} + \text{Gen}\, w_n \Delta\, \varphi_1(\kappa_n \Delta), \\ \int_{t_{j-1}}^{t_j} m_n\, ds &= \Delta \Big[m_n(t_{j-1}) \varphi_1(\kappa_n \Delta) + \text{Gen}\, w_n \Delta\, \varphi_2(\kappa_n \Delta) \Big] \end{aligned} \tag{H21.2}$$

and the surviving total $X = M + M_{\text{out}}$ of Eq. (P21.44) obeys

$$X(t_j) = X(t_{j-1}) e^{-\lambda \Delta} + \text{Gen}\, \Delta\, \varphi_1(\lambda \Delta) \tag{H21.3}$$

Both φ_1 and φ_2 tend to 1 and 1/2 for $x \to 0$, so a run with $\lambda = 0$ needs no special case. Written naively, φ_2 loses digits for $x < 0.1$ because it subtracts nearly equal numbers. The pseudocode therefore uses the series $\sum_k (-x)^k/(k+1)!$ and $\sum_k (-x)^k/(k+2)!$ for $x < 0.5$ (14 terms) and the closed forms above that. The decay-part script compares both against 50-digit values at 400 points between 10^{-12} and 10^{3} with a largest relative error of 2.9×10^{-16}.

The identities (H21.2) follow from multiplying the modal ODE (P21.28) by $4\pi I_n$ and using $4\pi s_n I_n = \text{Gen}\, w_n$; the first is Eq. (P21.33) and the second is the interval integral of it. Eq. (H21.3) is the same recursion for the single mode $\Lambda = 0$ applied to Eq. (P21.44). This was checked here by substitution as well as by the script. These three identities are not rows of the B3 reconciliation, which is why they carry the word HELPER.

```
BLOCK R1  modal solution with decay and source        planned module: src/analytical/modal_source.rs
INPUT   Phase 1 series: mode_rate[n] = Lambda_n (1/s), weight[n] = w_n, n = 1..n_modes
        (from the existing ModalRelease);
        decay_constant; initial kernel-uniform inventory inv0 (mol, may be 0);
        source schedule: list of (start s, end s, gen in mol per s), constant on each interval
          [gen = kernel volume * source_kernel from block E];
        output times
OUTPUT  at each output time: inv, dec_total, rel_total, gen_total, out_now, F_B, R_over_B (diagnostic)
```

```
R1.1  Initial modal inventories: mode_inv[n] = inv0 * weight[n].                (P21.46) APPROVED
      why: the kernel-uniform start profile projects onto the modes with the v1.0 weights.
      FUTURE: for another start profile, project it with N_n and I_n (P21.27, P21.29).
R1.2  total_survivors = inv0 ; dec_total = 0 ; gen_total = 0 ; t = 0.
R1.3  Boundaries = sorted list of 0, every start and end of the source schedule, and every output time,
      without repeats.
      FOR each interval (t, t+dt) between neighbouring boundaries:
R1.3a   gen = the schedule value on this interval (0 where none is listed).
R1.3b   FOR each mode n:
            mode_kappa = mode_rate[n] + decay_constant ; arg_mode = mode_kappa * dt
            f1 = phi1(arg_mode) ; f2 = phi2(arg_mode)      (series or closed form, see above)
            area[n] = dt * ( mode_inv[n] * f1 + gen * weight[n] * dt * f2 )          (H21.2)
            mode_inv[n] = mode_inv[n] * exp(-arg_mode) + gen * weight[n] * dt * f1   (H21.2, P21.33)
        why: this is exact for a constant gen, so a stepwise power history carries no time error.
             A source that is switched off is the case gen = 0 (P21.34).
R1.3c   dec_total = dec_total + decay_constant * (sum over n of area[n]).           (P21.38, P21.41)
        gen_total = gen_total + gen * dt.                                           (P21.41)
        total_survivors = total_survivors * exp(-decay_constant * dt)
                          + gen * dt * phi1(decay_constant * dt).                   (H21.3, P21.44)
R1.4  At an output time:
        inv       = sum over n of mode_inv[n].                                      (P21.35)
        rel_total = inv0 + gen_total - inv - dec_total.                             (P21.39)
        out_now   = total_survivors - inv.                                          (P21.44)
        F_B       = out_now / total_survivors.                                      (P21.45)
      why: the direct series for the release, sum of Lambda_n * area[n], converges only like 1/N because
           the weights decay like 1/n^2. The balance form replaces the two slow sums by their exact
           limits inv0 and gen_total (P21.39).
R1.5  Diagnostics, never used for the release totals:
        flux_direct = sum over n of mode_rate[n] * mode_inv[n]            (P21.37)
        completeness_defect = 1 - sum over n of weight[n]
        IF |completeness_defect| exceeds a user tolerance THEN advise more modes.
      why: the sum of w_n is 1 exactly; the missing part measures how many modes were dropped.
```

Truncation. The mode count follows two checks.

1. The v1.0 rule keeps modes up to $\Lambda_{\max}$ and trusts times $t \geq 40/\Lambda_{\max}$ for the series of M (`min_valid_time`, line 377). This controls only the inventory series. The release series needs the balance form of step R1.4.
2. For a homogeneous sphere the weights are positive and the balance form converges like N^{-3}, the direct flux series like N^{-1} (Section 3.5.2 of the derivation). The check script finds order 2.94 for the balance form between 10 and 100 modes. For the layered particle the weights change sign (152 negative weights among the first 300 modes in the check case), so the rigorous tail bound by $(1 - \sum w_n)/\Lambda_{N+1}$ is lost. Update 2026-11-13: Section 3.5.3 of the derivation gives explicit bounds that need no sign information, Bound A (P21.106) with the mode-count rule (P21.110) and Bound B (P21.112) (PROVED under completeness (O14) and the classical min-max and Sturm oscillation theorems; DEMONSTRATED NUMERICALLY; conservative); the steps of this block do not use them yet, and the diagnostic and mode-count steps that would are proposed in Section 9.5. The completeness defect of step R1.5 is a diagnostic and gives no bound. Section 3.9 of the derivation observed orders of 2.1 and 4.0 for the two-layer case.

Limits. R1 assumes constant diffusivities, so it covers a constant-temperature run. Temperature-dependent cases use R2 or R3. R1 also stops at a kernel-uniform source: the activation source of step E5 is spread over all layers and needs the general projection.

5.2 Block R2: effective time and its exactness check

Origin. Temperature part, T5.

In plain words. If every layer diffusivity is the same function of temperature up to

a constant factor, one clock change turns the whole variable-temperature problem into the constant-temperature problem of v1.0. The new clock is the effective time $\tau(t) = \int_0^t \alpha(s)\, ds$, where α is the common scaling. Decay disappears from the clock change after a factor $e^{-\lambda t}$ is split off. The released fraction is then the v1.0 closed-form series evaluated at τ instead of t.

This is valid only when the layers really share their temperature dependence, and real data rarely do. The program therefore runs a check first and refuses the shortcut when the check fails. A refusal is an error message with the reason. The program does not fall back to the shortcut with a warning and does not quietly use a mean activation energy.

Table 5.2 Equations of block R2.

Equation	Statement	Status	Class
(P22.24), (P22.25)	shared scaling $D_i(T(t)) = \alpha(t) D_i^*$; α for a single shared Q	APPROVED	DERIVED NOT IMPLEMENTED
(P22.26)	$\tau(t) = \int_0^t \alpha(s)\, ds$	APPROVED	DERIVED NOT IMPLEMENTED
(P22.27) to (P22.31)	$c = e^{-\lambda t}\hat{c}(r, \tau(t))$; decay cancels; transformed source $\hat{S} = e^{\lambda t} S/\alpha$	APPROVED	DERIVED NOT IMPLEMENTED
(P22.32)	without the decay factor the decay rate in τ is λ/α	APPROVED	DERIVED NOT IMPLEMENTED
(P22.33) to (P22.35)	inventory, flux and release by parts	APPROVED	DERIVED NOT IMPLEMENTED
(P22.36), (P22.37)	modal solution with variable α; constant α and source	APPROVED	DERIVED NOT IMPLEMENTED
(H22.2)	structural test for condition E1	DESIGN (sufficient test, not in B3)	DERIVED NOT IMPLEMENTED

The conditions E1 to E5 of Section 5.3.6 of the derivation decide whether the shortcut is exact (they are conditions of the model and are not the steps E1 to E5 of block E). Only condition E1 depends on the data.

Table 5.3 The conditions of the effective-time solution and how each is tested.

Condition	Meaning	How the program decides	Outcome for the planned input
E1	one factor $\alpha(t)$ multiplies every layer	structural test of the laws, Eq. (H22.2)	depends on the data
E2	temperature uniform in r	the history is one scalar function of t	true by construction
E3	one decay constant for the nuclide	one nuclide per solve	true by construction; a source of any shape is allowed
E4	absorbing outer surface	the only surface condition of the model	true by construction
E5	no partition coefficients at interfaces	none in the model	true by construction

The structural test for E1 compares each layer with layer 1, term by term, after sorting the terms by Q:

$$Q_{i,m} = Q_{1,m} \ \text{ for all } m, \qquad \frac{D_{0,i,m}}{D_{0,1,m}} = \text{ratio}_i \ \text{ independent of } m \tag{H22.2}$$

If Eq. (H22.2) holds for every layer then $D_i(T) = \text{ratio}_i\, D_1(T)$ for all T, which is Eq. (P22.24) with $\alpha(t) = D_1(T(t))/D_1(T_{\text{ref}})$ and $D_i^* = D_i(T_{\text{ref}})$. A single-layer problem passes with any law. The test is sufficient. It is also what E1 needs in practice: if the ratios D_i/D_j change at all along the history, no single clock exists (B3: two histories with equal $\int D_i\, dt$ in every layer but a different order give $F = 0.011413$ and 0.009919).

```
BLOCK R2  effective time                                planned modules: thermal_intervals.rs,
       ↪  temperature_history.rs
INPUT   laws of the layers; history; settings; decay_constant; output times
OUTPUT  Exact{ratio, T_ref, alpha_law} or NotExact{reasons}; then tau(t) and the solution       [NEW]

R2.1  check_effective_time(laws, history, settings)
      1. reasons = [].
      2. IF the number of layers is 1 THEN ratio = [1] and go to step 6.
      3. ref = layer 1 terms sorted by Q.
         FOR each layer i = 2..5:
             cur = layer i terms sorted by Q.
             IF len(cur) != len(ref) THEN add reason "layer i has a different number of terms".
             FOR each term m:
                 IF | Q_cur[m] - Q_ref[m] | > 1e-12 * max(Q_ref[m], 1) THEN add reason "layer i, term m:
       ↪  activation energies differ".
                  ratio_term[m] = D0_cur[m] / D0_ref[m]
              IF max(ratio_term) - min(ratio_term) > 1e-12 * max(ratio_term) THEN add reason "layer i: D0
       ↪  ratio depends on the term".
              ratio[i] = ratio_term[1]
        why: equality of floats read from files needs a tolerance; 1e-12 is far below any difference
             between two published correlations.
      4. IF any fluence rule is present and Phi_f is not frozen for the whole history THEN add reason
       ↪  "D0 changes in time".
        why: the fluence factor of (P22.9) acts on one term of one layer.
      5. Require the surface to be absorbing, no partition coefficients and a single decay_constant.
      6. IF reasons is not empty THEN return NotExact{reasons}.
      7. T_ref = the highest temperature of the history.                                    [DESIGN,
       ↪  decision 13]
        why: then alpha <= 1 and tau <= t; any T_ref is exact, this one keeps the series cheap to
       ↪  truncate.
      8. D_star[i] = diffusivity(law_i, T_ref) ;  alpha(T) = D_1(T) / D_1(T_ref).          (P22.24)
       ↪  APPROVED
         (For a single shared Q and one branch this is exp[-Q/Rg (1/T - 1/T_ref)], Eq. (P22.25).)
      9. RETURN Exact{...}.
```

Table 5.4 Outcome of the check for three inputs.

Input	Layers	Result	Reason
Real TECDOC-978 Cs data, FRG: kernel (209 and 362 kJ mol^{-1}), buffer (constant), PyC (222), SiC (125 and 514), PyC	5	NotExact	layer 2 has one term and layer 1 has two; the buffer has $Q = 0$ next to $Q > 0$
Synthetic data with one shared $Q = 300$ kJ mol^{-1} and different D_0 per layer	5	Exact	Eq. (H22.2) holds
Bare kernel, two-term Cs law (CRP-6 cases 1a and 1b)	1	Exact	one layer

The ratio $D_{\text{SiC}}/D_{\text{kernel}}$ of the real data varies by a factor 6.3 between 1250 and 1800 °C (the ratio is not monotonic: 7.2×10^{-4} at 1250 °C, a minimum of 4.2×10^{-4} near 1438 °C and 2.6×10^{-3} at 1800 °C, so the factor 6.3 is the largest over the smallest value; the integration check, whose kernel and SiC laws are these TECDOC-978 laws and whose buffer and PyC are synthetic, prints only the two end values, a factor 3.7; B7 review), so the failure is not marginal. Consequence for the project: with real data the effective-time solution applies to single-layer cases only and serves as a verification case with synthetic shared-Q data in the multilayer case (Section 5.3.6 of the derivation). The integration script reproduces the three outcomes of Table 5.4.

```
R2.2  tau_increment(law_1, T_ref, history, t_a, t_b)
      -- the same piecewise rule as step H1.4, applied to alpha = D_1(T) / D_1(T_ref)
      1. require [t_a, t_b] inside one segment.
      2. IF the slope is 0 THEN return (t_b - t_a) * D_1(T) / D_1(T_ref).
      3. otherwise sum over terms m, with arg_branch = (Q_m / Rg) * |1/T_b - 1/T_a|:
           arg_branch <= 1: Gauss-Legendre, 8 points                              [DESIGN threshold,
        ↪  evidence below]
           arg_branch  > 1: closed form (1/s) * D0_m * [T E2(theta_m / T)] from T_a to T_b      (P22.22)
        ↪  APPROVED
         then divide by D_1(T_ref).
      why: on a long ramp the closed form is exact in one evaluation; on a short piece it subtracts
           two nearly equal numbers.
R2.3  tau_at(t) = tau at the last breakpoint before t, plus tau_increment from there to t.
        ↪  (P22.26) APPROVED
      -- the running sum over breakpoints is stored once; tau is strictly increasing, so it can be
        ↪  inverted by bisection.
```

Evidence for the 8-point rule and its threshold (Cs in SiC, $|\Delta \ln D|$ over the piece against a 40-digit reference): Gauss-Legendre relative error 1.0×10^{-15} at 0.05, 4.4×10^{-16} at 0.1 and 0.5, 4.8×10^{-16} at 2, 1.9×10^{-13} at 4, 5.3×10^{-9} at 8 and 1.4×10^{-6} at 16. The closed form stays at 2×10^{-15} or better from 0.5 upward, and on four whole ramps (including the -600 K h^{-1} cool-down) it agrees with adaptive quadrature to 2.4×10^{-15} at worst. The threshold of 1 per term is therefore conservative.

```
R2.4  Evaluate the solution when R2.1 returned Exact
      1. layers = [(R_i, D_star[i])] ; times_tau = [tau_at(t) for t in output_times].
      2. decay_constant == 0 and no source:
           F(t) = F_v1(tau(t)), computed by the existing v1.0 routine exact_release on times_tau
        ↪  (P22.33) APPROVED
           (src/analytical/sturm_liouville.rs, lines 399 to 418). Points earlier than the validity time
           return "not valid" and no number (the v1.0 behaviour is kept).
      3. decay_constant > 0, no source:
           inventory   M(t)/M0 = exp(-decay_constant t) * sum_n w_n exp(-Lambda_star[n] * tau(t))
        ↪  (P22.33) APPROVED
           release     M_rel(t) = exp(-decay_constant t) * Mhat_rel(tau(t))
                         + decay_constant * integral_0^t exp(-decay_constant s) Mhat_rel(tau(s)) ds
        ↪  (P22.35) APPROVED
           the integral by Gauss-Legendre on each piece between breakpoints, using tau_at at the nodes.
           why: with decay the release depends on the whole path tau(s), s <= t, so one quadrature is
                unavoidable; it uses inventories only, which converge faster than the flux series.
           [Accuracy note, updated 2026-11-13 from Section 5.3.7 of the derivation; the step is
        ↪  unchanged.]
               The quadrature also needs Mhat_rel(tau(s)) for s earlier than the validity time of step 5,
                 where the truncated series is not valid, and for a bare kernel with a uniform start
        ↪  Mhat_rel
                 grows like sqrt(tau) near s = 0, so Gauss-Legendre converges only slowly on the first
        ↪  piece.
               Correction to the B7 note: the balance form M_rel = M0 - M - M_dec is the same expression
                 as (P22.35) (exact, PROVED, (P22.81)), so it has the same start problem and is not a
        ↪  separate
                 route; the earlier remark "(smooth integrand)" was wrong for the start piece.
                 For an absorbing surface and no source the start can be bounded (all PROVED):
                   modal start in closed form per mode on [0, t0], error at most
                  decay_constant * M0 * min(t0 * Wbar_N, Vbar_N / alpha_0) for constant alpha    (P22.93)
                   with the computable tail bounds Wbar_N, Vbar_N of (P22.90), (P22.91), and the bracket
                    (P22.94) for a varying alpha and c0 >= 0; for a bare kernel with a uniform start the
        ↪  image
                  form (P22.86) up to t_B, start error below 9.4e-20 M0 with 12 modes for constant alpha.
                 Late pieces with tau(b)/tau(a) <= 2 and 8 Gauss-Legendre points obey (P22.97) for
        ↪  constant
                 alpha per piece (CLAIMED ONLY on ramp pieces); the total (P22.98) is PROVED only when
        ↪  alpha is
                constant on the start piece and on every graded piece and under the coverage assumption
                 R24-A3c (late truncation (P22.95)). A layered particle with a long cold
                 start is OPEN (O17). The sub-steps that would apply these results are proposed in
        ↪  Section 9.5.
      4. With a source: each mode a_n(t) from the quadrature of Eq. (P22.36); the special case of constant
           alpha and constant source is Eq. (P22.37).
        ↪  (P22.36, P22.37) APPROVED
         The modal source projection comes from block R1.
      5. Series validity: require Lambda_covered_star * tau(t) >= 40 at every reported time (the v1.0
        ↪  rule
```

```
        min_valid_time applied to tau). Otherwise report "not valid".
```

Not covered by an approved equation (OPEN): a parent and daughter pair with different diffusivities, because the source of the daughter is then the parent field and the two fields do not share one clock. Such cases use the finite-volume path only.

Update 2026-11-13 (Section 5.3.8 of the derivation; the block is unchanged). A joint effective time for the pair exists only under conditions N1 and N2 of Theorem 1 (PROVED within the class (P22.101)): equal activation energies are necessary and not sufficient, and no linear ramp with a non-zero activation energy satisfies N2 when the decay coupling is present (Corollary 2). For identical diffusivities in every layer the pair has the closed form (P22.109) in τ, and whenever each species passes R2.1 on its own the daughter has an exact two-clock cascade (P22.113), (P22.114) whose truncation bound is OPEN. A chain mode of R2.1, a step 6 of R2.4 and a new block R2.5 that would use these results are proposed in Section 9.5; until the review accepts them, the finite-volume path stays the only route for such pairs.

5.3 Block R3: piecewise-constant temperature with eigenfunction projection

Origin. Temperature part, T6.

In plain words. When the temperature is constant on each interval, the problem on each interval is the v1.0 problem with that interval's diffusivities and it has its own exact eigenfunction series. At a switch the concentration profile is continuous in time and only its coefficients change, because the new eigenfunctions differ from the old ones. The program expands the old profile in the new eigenfunctions (a projection) and carries on. For a series that is not truncated, this is exact. This block gives the reference answer for CRP-6 case 3c and for any step schedule. If the layers share their temperature dependence (R2.1 passes), the eigenfunctions do not change and the projection matrix is the identity, so the program skips the projection and uses block R2.

Table 5.5 Equations of block R3.

Equation	Statement	Status	Class
(P22.39), (P22.40)	constant T on each interval; modal solution per interval	APPROVED	DERIVED NOT IMPLEMENTED
(P22.41) to (P22.43)	continuity of c at a switch; projection matrix P_{mn}	APPROVED	DERIVED NOT IMPLEMENTED
(P22.44)	mode-count rule $\Lambda_{n_{\max}+1}(t_k - t_{k-1}) \geq 40$	APPROVED (conservative)	DERIVED NOT IMPLEMENTED
(P22.45)	inventory mismatch ΔM_k at a switch as a diagnostic	APPROVED	DERIVED NOT IMPLEMENTED
(P22.46)	decay of the new coefficients like m^{-2}	approximate asymptotic estimate	DERIVED NOT IMPLEMENTED

```
Interval  = { t_start, t_end, T, D[1..5], Lambda_cover, n_modes }
Basis     = { Lambda[n], N[n], I[n], I1[n], shape_of_eigenfunction_in_each_layer }
            -- N[n] = int r^2 phi^2 dr (norm), I[n] = int r^2 phi dr, I1[n] = int_0^{R_1} r^2 phi dr
```

```
BLOCK R3  piecewise-constant temperature              planned modules: thermal_intervals.rs,
       ↪  modal_source.rs
INPUT   HistoryConstant; laws; radii; nuclide data; source schedule; output times
OUTPUT  M, M_dec, M_gen, M_rel, M_out at the output times; Delta_M_k at every switch              [NEW]

R3.1  prepare(history_constant, laws, radii, nuclide data)
      1. FOR each interval k: D_k[i] = diffusivity(law_i, T_k) (step C3); cache the basis by the exact
         value of the vector D_k.
         why: schedules often return to an earlier temperature; the eigen-solve is the costly part.
```

```
      2. Mode count per interval (DESIGN rule built from the approved criteria):
           the old-basis rule  Lambda[n_max + 1] * (t_end - t_start) >= 40                  (P22.44)
        ↪  APPROVED
           the new-basis rule  the series is valid from t_valid = t_start + 40 / Lambda_cover
         so Lambda_cover[k] = max( 40 / (t_end[k] - t_start[k]), 40 / (earliest requested output offset
        ↪  in interval k) ),
          capped by max_modes. Outputs earlier than t_valid are reported "not valid".
         why: this is the v1.0 'min_valid_time' idea applied per interval. Rule (P22.44) is conservative:
              the t_s = 1 h test with n_max = 10 had Lambda_11 * t_s = 32 < 40 and was already at
        ↪  reference accuracy.
R3.2  basis(D_k)            -- eigenpairs of one interval
      1. eigenvalues Lambda[n] up to Lambda_cover by the v1.0 shooting and Sturm-count machinery
         (eigenvalue_count and eigenvalues in sturm_liouville.rs).
      2. FOR each Lambda[n]: shoot from the centre to get u = r*phi in each layer as a cosine and sine
        ↪  pair,
         then N[n], I[n], I1[n] by the v1.0 panel rule (about one panel per quarter wavelength, 16-point
        ↪  Gauss-Legendre).
      NEEDS A DECISION (decision 12): the routine 'shoot' and the layer states are private in v1.0
         (line 132 of src/analytical/sturm_liouville.rs at tag v1.0.0). Either new
         public accessors are added to sturm_liouville.rs, or the shooting is copied into
        ↪  thermal_intervals.rs.
R3.3  start (k = 1)
      a[n] = c0 * I1[n] / N[n]   for a uniform kernel concentration c0.                (P21.29, P21.30)
      (this is the v1.0 weight ingredient: w_n = 3 I1[n] I[n] / (R_1^3 N[n]) per unit M_0)   (P21.46)
R3.4  propagate_inside_interval(a, Lambda, decay_constant, gen_k, t_from, t_to)
      FOR each mode n with mode_kappa = Lambda[n] + decay_constant:
          constant modal source s on the sub-interval of length Delta:
          a_new = a * exp(-mode_kappa * Delta) + (s / mode_kappa) * (1 - exp(-mode_kappa * Delta))
        ↪  (P22.40, P21.33) APPROVED
      why: exact in time, so a long interval needs no time stepping.
      Also accumulate the time integral of a_n over the sub-interval, for the decayed inventory:
          int a dt = a * (1 - exp(-mode_kappa Delta)) / mode_kappa
                     + (s / mode_kappa) * (Delta - (1 - exp(-mode_kappa Delta)) / mode_kappa)
        ↪  (P21.38) APPROVED
      (The same recursion in the stable form (H21.2) is used by block R1.)
R3.5  switch(a_old, basis_old, basis_new)             -- at t_k, k >= 2
      1. P[m][n] = ( 1 / N_new[m] ) * integral over r of u_new_m(r) * u_old_n(r) dr, layer by layer.
        ↪  (P22.43) APPROVED
         why: both bases are orthogonal with weight r^2, which does not depend on D when there are no
              partition coefficients. The products of two sines or cosines are integrated with the same
              panel rule as R3.2, with the panel count set by the sum of the two wave numbers (this
              avoids the singular closed form when two wave numbers coincide).                [DESIGN]
      2. a_new[m] = sum over n of P[m][n] * a_old[n].                                       (P22.42)
        ↪  APPROVED
      3. IF D_k == D_(k-1) in every layer THEN P is the identity; skip steps 1 and 2.
      4. Diagnostic: Delta_M = 4 pi ( sum_m a_new[m] I_new[m] - sum_n a_old[n] I_old[n] )     (P22.45)
        ↪  APPROVED
         Report it; it is zero for the infinite series. It measures the error right after the switch,
         not later: in the B3 check it was 3.0e-6 of M_0 at the switch while F later agreed to 6e-10.
R3.6  outputs at time t in interval k
      1. M(t) = 4 pi sum_n a_n(t) I_n                                                        (P21.35)
        ↪  APPROVED
      2. M_dec(t) = decay_constant * (accumulated integrals of M over all earlier sub-intervals plus the
        ↪  current one)   (P21.38, P21.41)
      3. M_gen(t) from the source schedule (known in closed form per piece).
      4. M_rel(t) = M_0 + M_gen(t) - M(t) - M_dec(t)  and  F = M_rel / M_0  (decay_constant = 0, no
        ↪  source).   (P21.39)
         why: the balance form. The flux series 4 pi sum Lambda I a converges like 1/m near a switch
              because the new coefficients fall only like m^-2 (about; observed m^-1.74 to m^-2.03), so
              the flux series must never be used just after a switch.                        (P22.46)
      5. Atoms outside the particle (decay_constant > 0):
         M_out(t) = M_0 exp(-decay_constant t) + integral_0^t gen(s) exp(-decay_constant (t - s)) ds -
        ↪  M(t)   (P21.44) APPROVED
         with gen the generation rate of the source schedule. No flux series is needed.
```

Truncation and accuracy notes. Coefficients of the new series fall like m^{-2} in order of magnitude when the layer diffusivities change by different factors (Eq. (P22.46), an approximate asymptotic estimate; the observed exponent depends on the case). The reconstructed profile shows Gibbs-type oscillation near interfaces right after a switch, so only integral quantities (inventories) are reported near a switch. Completeness of the layered eigenfunctions, which makes the projection exact for the infinite series, was closed in B3 by a Green-function argument (Hilbert-Schmidt); no textbook

reference was opened for it.

5.4 Block R4: staircase approximation of a ramp (verification tool only)

Origin. Temperature part, T7.

To test the finite-volume solver on a ramp with layer-specific activation energies, the ramp can be replaced by many short constant-temperature intervals and solved with block R3. The error shrinks as the intervals get shorter. This is slow (128 eigenvalue problems for an error below 10^{-6} on a 200 K ramp) and is not a production path.

Table 5.6 Equations of block R4.

Equation	Statement	Status	Class
(P22.53)	left, midpoint and per-layer average rules for the interval temperature	APPROVED	DERIVED NOT IMPLEMENTED
(P22.54), (P22.55)	Magnus terms and the defects of the three rules	APPROVED for small $h\|B\|$	DERIVED NOT IMPLEMENTED
(P22.56)	error orders 1, 2 and 2; observed order $p = \log_2(e_{N_T}/e_{2N_T})$	updated 2026-11-13 (Section 5.4.6 of the derivation): order 1 PROVED uniformly in the stiffness (P22.130); order 2 PROVED for shared scaling (P22.135) and, up to a factor $1 + \ln(T/h)$, on a fixed mesh (P22.140); with layer-specific Q and $h\|A\| \gg 1$ the V-norm order is about 1.75 in the test case and 1.51 to 1.96 in a second case (DEMONSTRATED NUMERICALLY); an order above 1 uniform in the mesh stiffness is OPEN	DERIVED NOT IMPLEMENTED
(P22.57)	with a shared scaling the average rule is exact at the interval ends	APPROVED	DERIVED NOT IMPLEMENTED

```
R4.1  staircase(linear segment [t_a, t_b], N_T, rule)       h = (t_b - t_a) / N_T
      FOR k = 0..N_T-1:
         rule LEFT:        T_k  = T(t_a + k h)                        -- error order 1
         rule MIDPOINT:    T_k  = T(t_a + (k + 1/2) h)               -- order 2 for F; on a fixed mesh up
        ↪  to 1+ln(T/h) (note below)
         rule AVERAGE:     Dbar_i,k = (1/h) * int D_i(T(t)) dt (per layer, as in step H1.4) -- order 2 as
        ↪  MIDPOINT      (P22.53) APPROVED
      -- the AVERAGE rule hands layer diffusivities, not a temperature, to block R3; so R3 must accept a
        ↪  list D[1..5] directly.
      Return a HistoryConstant (or a list of D vectors).
R4.2  observed order  p = log2( e(N_T) / e(2 N_T) )                                          (P22.56)
```

The orders 1, 2 and 2 are shown numerically for one stiff test problem (ramp 1600 to 1800 °C in 4 h and then a 2 h hold, synthetic layer-specific Q; last-pair orders 1.00, 2.03 and 1.95; B3 found 1.00, 2.01 and 2.04). A proof for the stiff regime ($h\|B\|$ of order 10^2 to 10^3) is OPEN; the Magnus argument holds for small $h\|B\|$. With a shared scaling factor the AVERAGE rule is exact at the interval ends (Eq. (P22.57)).

Update 2026-11-13 (Section 5.4.6 of the derivation; the steps above are unchanged). Order 1 of all three rules is PROVED uniformly in the stiffness, the mesh and λ (Theorem A, (P22.130)), and the LEFT rule cannot do better. Temperature jumps on the interval ends are exact and a jump inside an interval costs order 1 (Theorem B). With

shared scaling MIDPOINT has order 2 uniformly and AVERAGE is exact (Theorem C). With layer-specific Q order 2, up to a factor $1 + \ln(T/h)$, is PROVED only on a fixed mesh, with a constant that grows under mesh refinement (Theorem D, (P22.140)); in the V-norm of the concentration vector the observed order is 1.76 to 1.81 in the test case (1.75 on the finest mesh) and 1.51 to 1.96 in a second synthetic case, while the release fraction F keeps orders 1.98 to 2.13 (DEMONSTRATED NUMERICALLY). Consequence for step R4.2 and case VT18: measure the observed order on a smooth functional such as F, as VT18 does; a V-norm acceptance test can only require an order of at least 1, and a fixed band such as 1.75 ± 0.1 is not safe. A sub-step R4.0 that places interval ends on every kink and jump of $T(t)$ is proposed in Section 9.5; for a linear segment, as in R4.1, the segment ends already are the kinks.

5.5 Methods that are not planned

- **Galerkin system on a fixed basis** (Eqs. (P22.47) to (P22.51)): slow algebraic convergence when the diffusivity ratios change; not recommended in Section 5.4.5 of the derivation.
- **Proper orthogonal decomposition** (Eq. (P22.52)): the affine split in the layer diffusivities is inexact at the two interface faces because of the harmonic conductance; OPEN, not implemented, not tested.
- **Robin outer surface** (Eq. (P22.38)): v1.0 and this page use the absorbing surface.
- **Radial temperature profile or temperature-dependent partition coefficients**: outside the model (assumptions A1 and A2).

6 Glossary of variables

Units are SI. “Block” names the block where the name is first used. Local loop counters and temporary values inside one step are not listed.

6.1 Time and step control

Table 6.1 Time and step control.

Name	Symbol	Unit	Meaning	Block
`t`, `t_old`, `t_new`	t	s	time; start and end of the current step	M, F
`t_end`	t_{end}	s	end time of the run	A
`dt`, `dt_n`	Δt, Δt_n	s	step length, computed once as `t_new - t_old`	F
`dt_request`		s	requested step of the FIXED grid (v1.0 `dt_s`)	A, F
`n_steps`		1	number of steps of the FIXED grid	F
`dt_max`, `dt_first`		s	largest step; first step and first step after a switch or jump	F
`dt_decay`, `dt_diff`, `dt_T`, `dt_cap`		s	step limits from decay (P21.77), slowest diffusion mode (P21.78), temperature slope (P22.20); smallest fixed cap	F
`growth`		1	largest ratio of a step to the previous one (default 2.0)	F
`grow_cap`		s	limit on the next step from the growth rule	F
`eps_lambda`, `eps_1`, `eps_D`	ε_λ, ε_1, ε_D	1	allowed relative decay error, diffusion error and change of $\ln D$ per step	F

Name	Symbol	Unit	Meaning	Block
`events`			sorted list of (time, flags) with flags BREAK, JUMP, SWITCH, OUTPUT, END	F
`event_pointer`, `t_event`		1, s	position in the list and the next event time	F
`step_limit`		1	largest number of steps the user accepts	A, F
`output_every_steps`		1	v1.0 output interval in steps (FIXED grid)	A

6.2 Geometry and mesh

Table 6.2 Geometry and mesh.

Name	Symbol	Unit	Meaning	Block
`N`	N	1	number of cells	G
`R1`	R_1	m	outer radius of the kernel (region 1)	G
`R`	R	m	outer radius of the particle	G
`left`, `right`, `centroid[j]`	$r_{j-1/2}$, $r_{j+1/2}$, $\bar{r}_j$	m	edges and volume centroid of cell j	G
`volume[j]`	V_j	m^3	volume of shell cell j	G
`area[f]`, `area_R`	A_f, $4\pi R^2$	m^2	area of an internal face; area of the outer face	G
`dist_left[f]`, `dist_right[f]`, `dist_R`	δ_f^-, δ_f^+	m	distances from the face to the centroids on each side; outer distance $R - \bar{r}_N$	G
`cell_factor[j]`		1	fraction of cell j that lies inside the kernel (P21.54)	G
`layer[j]`	$i(j)$	1	layer index of cell j	G
`west_g[j]`, `east_g[j]`	g_j^W, g_j^E	$\mathrm{m}^3\ \mathrm{s}^{-1}$	diffusion conductance of the inner and outer face of cell j	H
`G_R`	G_R	$\mathrm{m}^3\ \mathrm{s}^{-1}$	surface conductance, `east_g` of the last cell; changes with D_5	H

6.3 State, decay and source

Table 6.3 State, decay and source.

Name	Symbol	Unit	Meaning	Block
`conc[j]`, `conc_old`, `conc_new`	C_j	$\mathrm{mol\ m}^{-3}$	cell-average concentration	H
`decay_constant`	λ	s^{-1}	probability per second that one atom decays	B
`half_life_s`, `mean_life_s`	$t_{1/2}$, $1/\lambda$	s	half-life and mean life	B
`z`	$z = \lambda \Delta t$	1	decay per step	H
`fission_rate_i`	$\dot{F}$	fissions $\mathrm{m}^{-3}\ \mathrm{s}^{-1}$	fission rate density in the kernel on interval i	E
`yield`	y	atoms per fission	cumulative yield used for the source	E
`f_q`, `y_q`	f_q, y_q	1, atoms per fission	fraction of fissile nuclide q and its yield	E
`N_A`	N_A	mol^{-1}	Avogadro constant, $6.02214076 \times 10^{23}$; BIPM, 2019, Table 1, p. 124	E

Name	Symbol	Unit	Meaning	Block
source_kernel_i	S_K	mol m^{-3} s^{-1}	source in the kernel on interval i	E
source_step	$\bar{S}$	mol m^{-3} s^{-1}	kernel source averaged over one step	H
source_cell[j]	$\bar{S}_j$	mol m^{-3} s^{-1}	cell-average source over one step	H
gen_rate	Gen	mol s^{-1}	generation rate in the particle, $\sum_j V_j \bar{S}_j$	H
t99	t_{99}	s	precursor delay $\ln 100/\lambda_p$ (P21.12)	E

6.4 Temperature and diffusivity

Table 6.4 Temperature and diffusivity.

Name	Symbol	Unit	Meaning	Block
T, T_old, T_end	T	K	temperature; at the start and at the end of a step	D, F, H
T_ref	T_{ref}	K	reference temperature of the effective time	R2
D0	D_0	m^2 s^{-1}	pre-exponential factor of one branch	C
Q	Q	J mol^{-1}	activation energy of one branch	C
Rg	R_g	J mol^{-1} K^{-1}	gas constant, 8.31446261815324 (SI) or 8.3143	C
Phi_f	Φ_f	10^{25} m^{-2}	frozen fast fluence for the SiC fluence factor (P22.9)	C
D_layer[i]	D_i	m^2 s^{-1}	diffusivity of layer i in the current step	H
D_floor		m^2 s^{-1}	smallest diffusivity the program accepts	C
Q_eff, Qmax	Q_{eff}	J mol^{-1}	apparent activation energy (P22.6); largest over layers	C, F
times_s, temps_K	t_k, T_k	s, K	breakpoints of a history	D
s	$\dot{T}$	K s^{-1}	slope of $T(t)$ on a segment	D, F
alpha	$\alpha(t)$	1	common scaling of all layer diffusivities (P22.24)	R2
tau	$\tau(t)$	s	effective time (P22.26)	R2
ratio_i		1	$D_{0,i,m}/D_{0,1,m}$ of Eq. (H22.2)	R2
D_star[i]	D_i^*	m^2 s^{-1}	diffusivity of layer i at T_ref	R2

6.5 Bookkeeping and fractions

Table 6.5 Bookkeeping and fractions.

Name	Symbol	Unit	Meaning	Block
inv, inv0	M, M_0	mol	inventory inside the particle now and at the start	J
gen_total	M_{gen}	mol	cumulative generated	J
dec_total	M_{dec}	mol	cumulative decayed inside	J
rel_total	M_{rel}	mol	cumulative released through the surface	J
out_now	M_{out}	mol	released and still undecayed outside	J
dec_out_total	$M_{\mathrm{dec,out}}$	mol	released and decayed outside	J
release_rate	J_R	mol s^{-1}	release rate $G_R C_N$	J
res_inside, res_outside		mol	closure residuals (P21.72), (P21.75)	J
F_rel, F_corr, F_gen, R_over_B, F_B, F_post		1	release fractions of Eq. (P21.45)	J
t_irr	t_{irr}	s	switch-off time of the source	J

6.6 Modal, projection and staircase

Table 6.6 Modal, projection and staircase names.

Name	Symbol	Unit	Meaning	Block
`mode_rate[n]`	Λ_n	s^{-1}	eigenvalue of mode n	R1
`weight[n]`	w_n	1	release weight of mode n per unit initial inventory	R1
`mode_kappa`	$\kappa_n = \Lambda_n + \lambda$	s^{-1}	mode rate with decay	R1
`arg_mode`	$x = \kappa_n \Delta$	1	argument of the interval factors	R1
`phi1(x)`, `phi2(x)`	φ_1, φ_2	1	interval factors of Eq. (H21.1)	R1
`mode_inv[n]`	$m_n = 4\pi I_n a_n$	mol	inventory carried by mode n	R1
`gen`	Gen	mol s^{-1}	generation rate on one interval of the schedule	R1
`total_survivors`	$X = M + M_{\mathrm{out}}$	mol	all atoms of the nuclide that have not decayed	R1
`N[n]`, `I[n]`, `I1[n]`	N_n, I_n, $I_n^{(1)}$	m^3 times eigenfunction scale	norm and moments of the eigenfunction	R3
`P[m][n]`	P_{mn}	1	projection matrix at a temperature switch (P22.43)	R3
`Delta_M`	ΔM_k	mol	inventory mismatch at a switch (P22.45)	R3
`N_T`, `h`	N_T, h	1, s	number and length of staircase intervals	R4
`arg_branch`		1	$(Q/R_g)\lvert 1/T_{\mathrm{end}} - 1/T_{\mathrm{old}}\rvert$, the change of $\ln D$ of one branch over a step	H, R2

6.7 Input fields, solver internals and temporaries

These names appear in the numbered steps and are listed so that every quantity of the pseudocode has a unit. Functions (for example `diffusivity`, `temperature_at`, `tau_at`) are not variables and are described in the block that defines them.

Table 6.7 Input fields, solver internals and temporaries.

Name	Symbol	Unit	Meaning	Block
`half_life_number`, `seconds_per_unit`		1, s	half-life as written in the data table and the seconds in one table unit	B
`precursor_half_life_s`, `warning_factor`	$t_{1/2,p}$	s, 1	half-life of the longest precursor; fraction of the time to the first output above which the precursor warning appears (default 0.1)	E
`source_history`		fissions $\mathrm{m}^{-3}\ \mathrm{s}^{-1}$	list of (start, end, fission rate density), constant on each interval	E
`output_times`		s	times at which a row is written	A, F
`q_unit`, `cutoff_tag`, `scale_1e25_per_m2`		text, text, $10^{25}\ \mathrm{m}^{-2}$	unit tag of Q; energy-cutoff tag of the fluence; fluence scale 5.0 of Eq. (P22.9)	C
`starts_s`, `end_s`	t_k	s	start times and end time of a constant-interval history	D
`diag_excess[j]`	v_j	$\mathrm{m}^3\ \mathrm{s}^{-1}$	volume term of the matrix diagonal, $V_j(1/\Delta t + \lambda)$ (P21.59)	H

Name	Symbol	Unit	Meaning	Block
`pivot_e[j]`, `den[j]`, `carry[j]`, `partial[j]`	e_j, d_j	$m^3\ s^{-1}$, $m^3\ s^{-1}$, $mol\ s^{-1}$, $mol\ m^{-3}$	elimination pivot; its sum with the east conductance; the term carried from the previous cell into the right-hand side; the forward-sweep value of the concentration (P21.61)	H
`rem`		s	time left to the next event during the landing loop	F
`Dbar_m`, `logs`, `top`	$\bar{D}_m$	$m^2\ s^{-1}$, 1, 1	step-average diffusivity of one branch; natural logarithms of the branch terms and their maximum	C, H
`out_exp`, `out_exp_old`		mol	optional exponential-form M_{out} of Eq. (P21.74)	J
`rel_total_at_t_irr`, `inv_at_t_irr`		mol	stored values at the step that ends on t_{irr}, for F_{post}	J
`flux_direct`, `completeness_defect`		$mol\ s^{-1}$, 1	diagnostic flux series (P21.37); $1-\sum_n w_n$	R1
`area[n]` (block R1)		mol s	time integral of the modal inventory over one interval (H21.2)	R1
`f1`, `f2`	φ_1, φ_2	1	values of the interval factors at `arg_mode`	R1
`max_modes`, `n_modes`, `Lambda_cover`		1, 1, s^{-1}	mode-count cap; modes used; eigenvalue up to which the series is trusted	R3
`t_valid`	t_{valid}	s	first time at which a truncated series is trusted, $40/\Lambda_{\mathrm{cover}}$ after the start of the interval	R2, R3
`t_a`, `t_b`		s	start and end of one piece of a history segment	H, R2
`times_tau`		s	effective times of the output times	R2
`reasons`, `ratio_term[m]`		text, 1	list of reasons for a refusal; D_0 ratio of one term in the structural test	R2

7 Verification cases and pass criteria

The cases below use the v1.0 five-layer particle of the check scripts (radii 250, 350, 390, 425 and 465 μm; synthetic diffusivities 4×10^{-14}, 4×10^{-13}, 2×10^{-14}, 5×10^{-15} and 2×10^{-14} $m^2\ s^{-1}$; cells (20, 8, 4, 4, 4)) unless stated. The last column states what a Python transcription of the pseudocode showed. A value in that column belongs to the transcription and not to Rust; each case must be repeated on the Rust code. The prefix V marks the decay, source and inventory cases, VT the temperature cases and VI the cases that need the joined run.

7.1 Decay, source and inventory

Table 7.1 Verification cases for decay, source and inventory.

ID	Case and reference	Pass criterion	Result of the transcription
V1	$\lambda = 0$, $S = 0$ against the v1.0 step	all cell values bit-identical over 200 steps (0 ulp)	PASS, 0 ulp
V2	decay without diffusion; $C^n = (1+z)^{-n} C^0$ (P21.63), $e^{-\lambda t}$	identity to 10^{-12} relative over at most 300 steps; error against $e^{-\lambda t}$ within 2 % of $\frac{1}{2}\lambda^2 t \Delta t$ (P21.77)	PASS, 1.2×10^{-14}; 1.5012×10^{-2} against 1.5000×10^{-2}
V3	decay with diffusion, no source; (P21.63)	identity to 10^{-11}; F_{corr} approaches the $\lambda = 0$ fraction at observed order 0.8 to 1.2	PASS, 4.6×10^{-15}; order 1.00
V4	constant kernel source, no decay; (P21.82) to (P21.84)	$\lvert J_R/\text{Gen} - 1\rvert < 10^{-12}$; $M_{\text{ss}}/\text{Gen} \to \tau_{\text{res}}$ at order 1.9 to 2.1	PASS, 1.3×10^{-15}; errors 5.5×10^{-6}, 1.4×10^{-6}, 3.5×10^{-7}, orders 1.97, 1.99
V5	source with decay, fixed point; (P21.85): $R/B = 0.9684291531$ (synthetic D, $\lambda = 2 \times 10^{-8}$ s^{-1})	order 1.9 to 2.1 in h; value on the (20,8,4,4,4) mesh within 2×10^{-6}	PASS, 0.968428661; order 1.99
V6a	Booth sphere, steady R/B; (P21.86)	order 1.9 to 2.1 in h for $N = 100$, 200, 400	PASS; $x = 1$: 4.1×10^{-6}, 1.0×10^{-6}, 2.6×10^{-7}; $x = 100$: 9.9×10^{-4}, 2.5×10^{-4}, 6.2×10^{-5} (orders 2.00; $x = 100$ misses the 10^{-5} tolerance at $N = 200$)
V6b	Booth sphere, transient $F_B = M_{\text{out}}/(M + M_{\text{out}})$; (P21.87), TECDOC-1674 (7.8), $x = 1$	first order in Δt when Δt and the grid are halved	PASS; errors 5.2×10^{-4}, 2.7×10^{-4}, 1.4×10^{-4}, order 0.94
V7	modal series (R1) against the finite-volume run, five layers, source then decay	difference over $M_{\text{gen}}(t_{\text{irr}})$ below 2×10^{-3} at the finest mesh, factor 1.8 or more per halving	PASS, worst 9.7×10^{-4} at the finest mesh (80, 32, 16, 16, 16)
V8	manufactured solution, two layers; (P21.88) to (P21.90)	order within 0.1 of 2 (steady) and 1 (transient)	not in the transcription script; `sec4_p21_mms_source.py` passed in the B3 re-run
V9	closure with switching source and variable steps; (P21.72), (P21.75)	relative residual below 10^{-12}	PASS, 9.5×10^{-16} and 6.3×10^{-16}
V10	positivity; (P21.62)	minimum of C is ≥ 0.0 for $\Delta t = 10^{-3}$ to 10^{12} s	PASS, smallest value 1.2×10^{-127}
V11	CRP-6 case 1a (no source, no decay, one region); (P21.91) with $R_g = 8.3145$ in the transcription	$\lvert F - 0.46742\rvert < 2 \times 10^{-4}$	PASS, $F = 0.46737$ at $\Delta t = 360$ s, $N = 150$; step C2 offers only the SI value and 8.3143, whose series values 0.467404 and 0.467337 are also inside the tolerance (B7)
V12	CRP-6 case 1b; $1 - F$ from the series is 4.0703×10^{-7} with $R_g = 8.3145$ (4.0746×10^{-7} SI, 4.0936×10^{-7} with 8.3143; the Rust test takes the series value for the R_g it runs)	within 5 % at $\Delta t \leq 36$ s; the error at larger steps follows (P21.78)	PASS, 4.0903×10^{-7} (+0.5 %); at 360 s +5.14 % against 5.05 % predicted

ID	Case and reference	Pass criterion	Result of the transcription
V13	source switch inside a step	M_{gen} exact to 10^{-13} with option A	PASS, error 0; option R is off by 3×10^{-3}
V14	steady state preserved by Backward Euler	profile change below 10^{-11} for a step of any size	PASS, 9.1×10^{-16}
V15	decay-exact split as cross-check, $S = 0$; (P21.67)	equals $e^{-\lambda t}\times$ v1.0 run to 10^{-12}	PASS, 3.3×10^{-15}
V16	block B against `decay.csv`	seconds to 10^{-8}, λ to 5×10^{-7} relative; a stable row gives 0; a negative half-life is rejected	PASS, 2.4×10^{-10} and 3.9×10^{-7}
V17	cell and step averages of block E and G; (P21.54)	agreement to 10^{-12}	PASS, 4.4×10^{-16}
V18	block R1 recursion against ODE integration	relative difference below 10^{-8} (25 modes)	PASS, 8.2×10^{-16} and 6.4×10^{-15}
V19	block R1, Booth sphere	absolute difference below 5×10^{-9} against (7.8) at $x = 1$	PASS, ten printed digits (0.8731144969)
V20	block R1, $\lambda = 0$; (P21.76)	$M_{\text{out}} = M_{\text{rel}}$ to 10^{-12} relative	PASS
V21	helper identities (H21.1) to (H21.3)	symbolic residual 0; relative error below 10^{-14}	PASS, 2.9×10^{-16}

Which cases belong in `cargo test`: the algebraic cases V1, V2, V3 (identity part), V9, V10, V13, V14, V15, V16, V17 and V21 run in seconds on a small mesh. The accuracy cases V4 to V8, V11, V12 and V18 to V20 need a mesh and step study and fit `examples/verify.rs` or a longer test file.

Two differences between the transcription and Section 4.8 of the derivation need the reviewer's attention. Section 4.8 test 6 gives a steady Booth tolerance of 10^{-5} relative at $N = 200$; the transcription meets it at $x = 1$ and misses it at $x = 100$ because the thin boundary layer needs more cells, so the criterion of V6a is the observed order. And Section 4.8 test 7 named its reference as the Phase 2.2 modal solution, while the reference is the modal series of block R1, which belongs to the constant-temperature part; the B7 review corrected the derivation text to "Phase 2.1 modal solution of Section 3".

7.2 Temperature and diffusivity

Table 7.2 Verification cases for the Arrhenius and temperature-history blocks.

ID	What is tested	Pass criterion	Evidence now
VT1	input validation, steps A8, C1	each bad input returns an error that names the field; no panic; $Q = 0$ accepted	9 of 9 reference checks pass
VT2	$D(T)$ by log-sum-exp against a 40-digit evaluation (4 register laws, 5 temperatures)	relative difference $\leq 10^{-14}$; $\ln D$ finite where $e^{-Q/(R_g T)}$ underflows	worst 8.4×10^{-15}; $\ln D = -1097.91$
VT3	Q_{eff} and crossover	130.1 and 438.2 kJ mol^{-1}, 1772.4 K within 0.1	130.10, 438.22, 1772.44
VT4	gas constant, CRP-6 1a	0.467337 (8.3143) and 0.467404 (SI) within 2×10^{-6}	pass
VT5	history construction, step D1 to D3	AGR-1 totals 333.78, 335.95, 338.12 h and nine CRP-6 totals within 0.01 h; non-increasing times rejected	pass

ID	What is tested	Pass criterion	Evidence now
VT6	interpolation, step D5; (P22.21)	$T(t_k)$ equals the table value exactly; no extrapolation	pass
VT7	landing, step F5; (H22.1)	every event is a step end by float equality; a duplicate event leaves the step sequence and the final state bit-identical	41 of 41 events; sequence identical
VT8	step bound; (P22.20)	largest $\lvert\Delta \ln D_i\rvert \le 2\varepsilon_D$ in every step; no step below 1 s next to a step of 100 s	0.102 for $\varepsilon_D = 0.1$; shortest 1.24 s
VT9	effective-time integral, step R2.2	closed form $\le 10^{-12}$; Gauss-Legendre $\le 10^{-13}$ while the argument is at most 1	2.4×10^{-15} worst; $\le 1.4 \times 10^{-15}$
VT10	FV against the series at $\tau(t)$ (shared Q)	spatial order 2.0 ± 0.1; temporal order 1.0 ± 0.1	order 2.00, difference 5.4×10^{-7} at 160 cells per layer
VT11	option A against Backward Euler in τ ($\lambda = 0$, shared α)	relative difference $\le 10^{-11}$; for $\lambda > 0$ a first-order difference is expected	3.5×10^{-15}; 2.7×10^{-3} (B3); 4×10^{-16} in the integration check
VT12	exactness check refuses unshared data, step R2.1	NotExact for layer-specific Q; shared Q passes and the two orders agree to 10^{-13}	$F = 0.011413$ against 0.009919; shared Q 8.2×10^{-15}; the three outcomes of Table 5.4 reproduced
VT13	closure with a changing G_R; (P22.16)	relative residual $\le 10^{-12}$ at every step; the fixed-G_R booking of v1.0 fails (above 10^{-8})	4.8×10^{-15}; negative test 5.8×10^{-5} (`sec6_pseudocode_checks.py`); 1.2×10^{-5} in the integration check
VT14	positivity; (P22.17)	minimum concentration ≥ 0.0	0.0
VT15	decay with effective time ($\lambda > 0$); (P22.33), (P22.35)	inventory and release relative difference $\le 10^{-11}$ to a stiff real-time integration	1.6×10^{-12} and 2.8×10^{-12}
VT16	regression to v1.0 (constant T, single-term $Q = 0$ laws)	results bit-identical to v1.0	pass for 200 steps in Python
VT17	projection at a temperature step, block R3	agreement $\le 10^{-6}$ in F with a Richardson-extrapolated FV for a 1 h step; $P = I$ when the diffusivities do not change; ΔM_k reported	6.9×10^{-11}; 1.5×10^{-10} for a 0.05 h step
VT18	staircase orders, block R4	last-pair orders 1.0 ± 0.15, 2.0 ± 0.15, 2.0 ± 0.15 (left, midpoint, average) for $N_T \ge 32$	1.00, 2.03, 1.95 (B3: 1.00, 2.01, 2.04)

7.3 Cases that need the joined run

These cases test what neither part could test alone. They are new in this page and have a Python transcription in `b4_phase2_pseudocode_integration.py`.

Table 7.3 Verification cases for the joined run.

ID	What is tested	Pass criterion	Result of the transcription
VI1	planner landing with events from the source and from the history	every event time is a step end by float equality; the step after a switch is at most `dt_first`; no step above `dt_max` or `dt_decay`; largest change of $\ln D$ at most $2\varepsilon_D$	PASS; 8 of 8 events, 139 steps, largest $\lvert\Delta \ln D\rvert = 0.102$

ID	What is tested	Pass criterion	Result of the transcription
VI2	closure of the joined run (ramp, decay, source off at 3 h), options A and E	both residuals below 10^{-12}; minimum concentration at least 0.0; generated amount equal to its analytic integral to 10^{-13}	PASS; 1.3×10^{-15} (A), 1.2×10^{-15} (E); generated amount error 0
VI3	negative test, release booked with the first-step G_R	residual above 10^{-8}	PASS, 1.2×10^{-5}
VI4	joined loop in FIXED mode against the v1.0 transcription	every row and the final profile bit-identical	PASS, 200 steps
VI5	constant-interval history with a jump: left-limit rule of step D6	the jump is a step end; no step shorter than 1 s apart from the first; closure below 10^{-12}; option E on the step that ends at the jump uses the old temperature	PASS, 36 steps; the naive rule gives 66 steps and a shortest step of 5.6×10^{-7} s
VI6	option A against Backward Euler in τ over five layers with shared Q	relative difference $\leq 10^{-11}$	PASS, 4×10^{-16}

8 Cross-check of every step against the reconciled equations

8.1 Method

Every step that carries an equation number was compared with the display equation of that number in `sec3_phase21_decay_source_analytical.md`, `sec4_phase21_decay_source_fv.md` and `sec5_phase22_temperature.md`, with the reconciliation record `B3_reconciliation.md` as the arbiter. A script (`extract_eqs.py` in the work folder) pulled the 151 tagged display equations (P21.1 to P21.92 and P22.1 to P22.59) out of the three sections so that each cited number could be read next to the step that cites it. The comparison of meaning was done by reading, with the following quantities recomputed:

- the elimination recurrence of step H4.3 against Gaussian elimination of the matrix of Eq. (P21.58), and against the v1.0 code of `solve_diffusion_system` (lines 132 to 160);
- the interval formulas (H21.1) to (H21.3) against Eqs. (P21.33), (P21.38) and (P21.44);
- the identity $\int e^{-\theta/T} dT = T\, E_2(\theta/T)$ behind Eq. (P22.22);
- the step limits of Eqs. (P21.77), (P21.78) and (P22.20) with the example values;
- the v1.0 line numbers quoted in this page, against the tag.

8.2 Mismatches found and how the pseudocode was changed

No equation was changed. Each row below names a place where the drafts disagreed with each other, with the v1.0 code, or with the equation they cite.

Table 8.1 Mismatches and changes.

ID	Where	Mismatch	Change in the pseudocode
X1	decay part C1 against temperature part T3	two step planners with different growth factors (1.5 against 2.0), different handling of the last step before an event (common scale factor against halving the remainder), and a restart after an event in one and not in the other	merged into block F: landing by assignment and halving of the remainder, restart at `dt_first` after SWITCH and JUMP only, one `growth` (default 2.0), caps from both parts; decision 10
X2	decay part C1.5 against v1.0 lines 190 and 191	with growth 1 the geometric rule gives the v1.0 step only to round-off (the common scale factor is not the same arithmetic as `end_time / steps`; in 19314 of 20000 random pairs of end time and step the last bits differ, by at most 1.8×10^{-14} relative), but cases V1 and VT16 demand bit identity	step F0: FIXED mode with the v1.0 rule; block A refuses FIXED when an event time exists
X3	temperature part T2 (right-continuous constant kind) against T3 step 5 and T4.2 option E	`temperature_at(t_new)` at a jump returns the temperature of the next segment, so the step guard sees an impossible jump of $\ln D$ and halves the last step before every jump, and option E would use the new temperature for the old step. The check script reproduces it (66 steps, shortest step 5.6×10^{-7} s, against 36 steps)	new step D6 (`temperature_at_step_end`, left limit); used in F5.5 and H1.3
X4	decay part D4 against temperature part T4.5	D4 defined G_R as the east conductance of the last cell without saying that it must be the one of the current step; v1.0 line 210 reads the fixed mesh constant, which is wrong once D_5 changes	J1.2 states that G_R is the one used in the matrix of the same step; negative test VI3
X5	decay part Table 6A.11	the relation $4\pi s_n I_n = G\, w_n$ was attributed to (P21.46), (P21.47); (P21.47) is the R/B series	cited as (P21.30) with (P21.46) in Table 5.1
X6	temperature part T5.1 step 8	$\alpha(T) = D_1(T)/D_1(T_{\mathrm{ref}})$ was cited to (P22.24), (P22.25); (P22.25) holds only for one branch with a shared Q, and the general statement is in the text of Section 5.3.1	cited to (P22.24), with (P22.25) named as the single-branch case (R2.1 step 8)
X7	decay part C1.3	(P21.78) is derived for constant diffusivities, yet the step was written without a restriction	marked OPEN for a temperature history in F3 (use Λ_1 at the highest temperature as a warning only); update 2026-11-13: with shared scaling and option A that value is a PROVED sufficient cap (not proved for option E, whose lag L_1 enters), see the F3 note and (P22.69) to (P22.72)

ID	Where	Mismatch	Change in the pseudocode
X8	both parts	symbols used twice: G (generation and conductance), κ (mode rate and layer ratio), R (gas constant and radius), e (event and pivot), x (three arguments), B (birth rate and step matrix)	renamed (Table 1.3); B appears only in cited derivation text
X9	temperature part equations (6.1) and (6.2)	numbered locally and not in the reconciled sections	renamed (H22.1) and (H22.2), status DESIGN; the helpers (H21.1) to (H21.3) are HELPER
X10	both parts, verification	two ID series (V and VT) and an overlap between V9 and VT13	kept as two series; the overlap is noted in Table 7.2; new series VI for the joined run
X11	decay part, open item O6	Section 4 of the derivation refers to the v1.0 step at lines 171 to 180; in the tag it is lines 162 to 182 with the changed lines at 178 and 179	not a pseudocode change; carried as O6 in Section 9.3. Resolved in the B7 review: lines 171 to 180 are the assembly loop inside `backward_euler_step`, and Section 4.3 of the derivation now cites the function at lines 162 to 182 with that loop named
X12	temperature part D1 example	named two literature sources for the missing ramp rates by author	replaced by the register rows (TH-001 onward); no reference is cited that this page cannot show
X13	decay part, interaction paragraph and Table 6A.6, against this page	the decay part states that the decay factor of Eq. (P21.80) holds for a diffusivity that changes in time and cites (P21.53) for the cell average; this page cited neither, so nothing in block H said why decay and a changing D may be combined	cited (P21.80) in Table 4.10 and in the new paragraph “Decay and a changing diffusivity” (restriction: one constant λ); cited (P21.53) in Table 4.8

8.3 Status of the equations used

The word APPROVED in the tables of Section 4 and 5 means that the equation is in the reconciled Sections 3 to 5 and that either B3 re-derived it or the section check logs and the 17 section scripts that B3 re-ran cover it. The distinction matters, so the equations cited in Sections 4 and 5 that are not mentioned anywhere in `B3_reconciliation.md` are listed here (found by a script that reads the B3 text and expands its ranges). Many are definitions or one-line consequences of an equation that B3 did confirm; the Phase 2.2 finite-volume relations (P22.13) to (P22.23) are the largest group, and for them the support is the Section 5 check log, the scripts `sec5_*.py` and the closure and positivity runs of Section 7 of this page, not a separate B3 row.

```
P21.1 P21.5 P21.6 P21.7 P21.8 P21.9 P21.10 P21.11 P21.13 P21.15 P21.18 P21.23 P21.26 P21.27
P21.29 P21.30 P21.34 P21.35 P21.37 P21.38 P21.40 P21.41 P21.45 P21.46 P21.52 P21.54 P21.55
P21.66 P21.67 P21.71 P21.72 P21.79 P21.90 P21.92
P22.5 P22.9 P22.10 P22.13 P22.14 P22.15 P22.16 P22.17 P22.18 P22.19 P22.20 P22.21 P22.22
P22.23 P22.33 P22.37 P22.38 P22.39 P22.47 P22.51 P22.52 P22.53 P22.57 P22.58 P22.59
```

Steps that rest on OPEN items are marked in the text: E1 (precursor warning factor), E5 (activation products), C1.4 (fluence cutoff), D2.4 and D2.5 (ramp shape, row 8a), F3 (variable temperature), R3.2 (access to eigenfunctions), R4 (stiff regime), R2.4 (parent and daughter with different diffusivities) and R2.4 step 3 (start of the release

quadrature, added in the B7 review). Update 2026-11-13: F3 rests on an unproved bound for layer-specific Q (O12) and, under shared scaling, for option E; R4 only for an order above 1 that is uniform in the mesh stiffness (O10); the parent-daughter note after R2.4 for the truncation of the cascade and for transformations outside the class (P22.101) (O11); and R2.4 step 3 for a layered particle with a long cold start and for ramp pieces ((P22.87), (P22.97); O17).

B7 final review (2026-11-06). Changes to this page: the solution-path row for a ramp now says that no general closed form is available (Section 5.4.2 of the derivation); the worked scalar example of block H now calls its difference a one-step local error of order z^2; R2.4 step 3 carries the accuracy note on the start of the release quadrature; the remark under Table 7.1 and items X11 and O6 record the corrections made to Section 4 of the derivation; the closing sentence of Section 9.4 lists every new non-Rust file. No equation, step or verification result was changed.

Second integration pass (2026-11-06, after the decay part was rewritten). The rewritten decay part and the unchanged temperature part were compared with this page again. The numbers of the decay part (four decay constants, the Cs-137 source, the two step limits, the yield table, the scalar example, the verification cases V1 to V21) agree with this page, and the re-run of its script gives 55 of 55 PASS. Changes made on this page: item X13 above; the error values of cases V4, V6a and V6b and the mesh of V7 and V11 are now written in Table 7.1 instead of being left to the part draft; Table 6.7 lists the input fields, the solver internals and the temporaries of the numbered steps with their units. No equation, step or other verification result was changed. The pseudocode still carries the merged planner of block F (growth 2.0, landing by halving the remainder), not the growth factor 1.5 of the decay part; that is item X1 and decision 10.

9 Status, open items and decisions

9.1 Classification of the operations

Table 9.1 Classification of the operations. The last five rows were added on 2026-11-13; they classify results of the open-item work that no step uses yet.

Operation	Class	Reference
mesh, volumes, series conductances	IMPLEMENTED	src/numerical/finite_volume.rs, lines 50 to 99
subtraction-free solver	IMPLEMENTED	lines 132 to 160
inventory sum	IMPLEMENTED	lines 114 to 120
Kahan release sum and release rate	IMPLEMENTED	lines 209 to 217
modal eigenvalues and weights w_n	IMPLEMENTED	src/analytical/sturm_liouville.rs, lines 308 to 395
exact release from the series at given times	IMPLEMENTED	lines 399 to 418
decay constant, source, yield, cell and step averages (B, E, G4)	DERIVED NOT IMPLEMENTED	(P21.7), (P21.8), (P21.10), (P21.54), (P21.57)
step with decay and source (H4)	DERIVED NOT IMPLEMENTED	(P21.58), (P21.59)
events and planner (F)	DERIVED NOT IMPLEMENTED	(P21.77) to (P21.79), (P22.20); the landing and restart rules are DESIGN
bookkeeping with M_{gen}, M_{dec}, M_{out} and fractions (J)	DERIVED NOT IMPLEMENTED	(P21.45), (P21.72) to (P21.76), (P22.16)
Arrhenius diffusivity, apparent Q, crossover (C)	DERIVED NOT IMPLEMENTED	(P22.1), (P22.6), (P22.7)
histories (D)	DERIVED NOT IMPLEMENTED	(P22.21), (P22.39), (P22.58), (P22.59)

Operation	Class	Reference
conductances at the step temperature, step average (H1, H2)	DERIVED NOT IMPLEMENTED	(P22.14), (P22.22)
modal solution with decay and source (R1)	DERIVED NOT IMPLEMENTED	(P21.28) to (P21.39), (P21.44)
effective time (R2)	DERIVED NOT IMPLEMENTED	(P22.24) to (P22.37)
projection at a temperature switch (R3)	DERIVED NOT IMPLEMENTED	(P22.40) to (P22.46)
staircase of a ramp (R4)	DERIVED NOT IMPLEMENTED	(P22.53) to (P22.57)
activation products (E5)	FUTURE	(P21.13) to (P21.16), data NOT VERIFIED
parent-daughter chains with overlap matrix	FUTURE	(P21.17) to (P21.22)
projection of an arbitrary start profile onto the modes	FUTURE	(P21.27), (P21.29)
precursors as separate species	FUTURE	Section 3.3.4 of the derivation
Galerkin on a fixed basis, POD, Robin surface	FUTURE (not planned)	(P22.47) to (P22.52), (P22.38)
truncation bounds and mode count for R1 (proposed steps R1.0 and R1.5)	DERIVED NOT IMPLEMENTED; proposed in Section 9.5, not yet a step	(P21.105) to (P21.112)
diffusion step rule with a temperature history (proposed change of F3 to F5)	DERIVED NOT IMPLEMENTED; proposed in Section 9.5, not yet a step	(P22.69) to (P22.74)
bounded start of the release quadrature (proposed R2.4 sub-steps 3a to 3e)	DERIVED NOT IMPLEMENTED; proposed in Section 9.5, not yet a step	(P22.81) to (P22.99)
parent-daughter closed form and two-clock cascade (proposed R2.1 chain mode, R2.4 step 6, block R2.5)	DERIVED NOT IMPLEMENTED; proposed in Section 9.5, not yet a step	(P22.105) to (P22.114)
second-order or exponential integrator (proposed step H4b, decision 15)	DERIVED NOT IMPLEMENTED; proposed in Section 9.5, not yet a step	(P21.113) to (P21.142)

9.2 Status of the numerical properties

Table 9.2 Status of the numerical properties.

Property	Statement	Class	Basis
Positivity of the Backward Euler step	$C^{n+1} \geq 0$ for every $\Delta t > 0$, $\lambda \geq 0$, every temperature, $C^n \geq 0$, $S \geq 0$	PROVED	Eqs. (P21.62), (P22.17); sums, products and quotients of nonnegative numbers only; V10, VT14 and VI2 show no violation
Subtraction-free pivots stay valid with decay	$d_j = e_j + g_j^E$	PROVED	Eq. (P21.61) by induction
Inventory closure	$M_0 + M_{\text{gen}} = M + M_{\text{dec}} + M_{\text{rel}}$ per step and summed, also with a changing G_R	PROVED for the exact linear solve; DEMONSTRATED NUMERICALLY at round-off	Eqs. (P21.70), (P21.72), (P22.16); V9 gives 10^{-15}, VI2 gives 1.3×10^{-15}
Outside-inventory closure	$M_0 + M_{\text{gen}} = M + M_{\text{out}} + M_{\text{dec}} + M_{\text{dec,out}}$	PROVED; DEMONSTRATED NUMERICALLY	Eq. (P21.75); V9, VI2
Steady state independent of Δt	fixed point of the unsplit step equals the semi-discrete steady state	PROVED	Eq. (P21.68) discussion; V14 shows 10^{-15}

Property	Statement	Class	Basis
Time accuracy of Backward Euler for decay	first order; decay error $\frac{1}{2}\lambda^2 t\Delta t$	PROVED as a leading-order expansion; DEMONSTRATED NUMERICALLY	Eqs. (P21.64), (P21.77); V2 gives a 0.1 % match
Space accuracy	second order in h (steady and transient)	DEMONSTRATED NUMERICALLY	V4, V5, V6a give orders 1.97 to 2.00
Time accuracy of the joined scheme (decay, source and changing $D(T)$ together)	first order	CLAIMED ONLY	each part was shown separately (V3, V12, VT10); the joined run has been checked for closure and positivity (VI2) and has not been convergence-tested
Option A equals Backward Euler in τ	for a shared scaling factor and $\lambda = 0$	PROVED by substitution (every conductance equals $\Delta\tau_n/\Delta t_n$ times its reference value); DEMONSTRATED NUMERICALLY	Eq. (P22.57) and the text of Section 5.2.3; VT11 (3.5×10^{-15}), VI6 (4×10^{-16})
Effective-time solution exact	under E1 to E5	PROVED (symbolic, B3); DEMONSTRATED NUMERICALLY	Eqs. (P22.24) to (P22.33); VT10, VT15
Modal recursion exact in time	piecewise-constant source, no time error	PROVED	Eq. (P21.33); V18
Projection at a temperature switch	exact for the untruncated series	PROVED (completeness by the Green-function argument of B3); DEMONSTRATED NUMERICALLY	Eqs. (P22.40) to (P22.43); VT17
Staircase orders 1, 2, 2	left, midpoint, average	updated 2026-11-13: order 1 PROVED uniformly in the stiffness; order 2 PROVED for shared scaling (uniformly) and, up to a factor $1 + \ln(T/h)$, on a fixed mesh (constant grows like $N^{1/2}$, DEMONSTRATED NUMERICALLY); orders near 2 for F DEMONSTRATED NUMERICALLY; V-norm order 7/4 CLAIMED ONLY; an order above 1 uniform in the mesh stiffness for layer-specific Q OPEN	Eqs. (P22.54) to (P22.56), (P22.130), (P22.140); Section 5.4.6 of the derivation; VT18; `sec5_o10_stiff_staircase.py`
Gauss-Legendre average with 8 points	relative error below 10^{-15} while the change of $\ln D$ is at most 2	DEMONSTRATED NUMERICALLY	`b4_gl8_range.py`; VT9

Property	Statement	Class	Basis
Truncation of the balance form	N^{-3} worst case for sign-definite weights; updated 2026-11-13: a rigorous bound for the layered particle whose source and decay part falls like $(N-J)^{-3}$, and for $S=0$ and an absorbing surface a computable a posteriori tail bound	PROVED asymptotically for the homogeneous sphere; DEMONSTRATED NUMERICALLY otherwise; layered bound PROVED under completeness (O14) and the classical min-max and Sturm oscillation theorems; tail bound for $S=0$ and an absorbing surface PROVED	Section 3.5.2 of the derivation; layered weights change sign; (P21.106); (P22.90), (P22.91)
Rigorous truncation bound for the layered particle	updated 2026-11-13: Bound A (P21.106) with corollaries (P21.107) to (P21.110), Bound B (P21.112), for a kernel-uniform start and source	PROVED (conditional on completeness, O14, and on the classical min-max and Sturm oscillation theorems); DEMONSTRATED NUMERICALLY; conservative	Section 3.5.3 of the derivation; `sec3_o4_layered_truncation_bound.py`
Diffusion step limit with a temperature history	rule (P22.69) with step (P22.72) keeps the slow-mode error below ε_1 for option A; for option E the rule (P22.74) with ζ_E bounds it by $\varepsilon_1 + \frac{1}{2}\zeta_E^2\,\mathrm{TV}(\ln\Lambda_1)$; the F3 value with Λ_1 at the highest temperature is a sufficient cap for option A	PROVED for shared scaling (option A by (P22.69) with (P22.72), option E by (P22.74); the F3 cap for option A only); frozen-coefficient part PROVED for layer-specific Q (option A for any history, option E on heating only); remainder CLAIMED ONLY	Section 5.2.5 of the derivation; `sec5_o12_variable_D_step_limit.py`
Start of the release quadrature with decay	start bound (P22.93), bracket (P22.94) and total (P22.98), absorbing surface, $S=0$	(P22.93) PROVED for constant α; bracket (P22.94) PROVED for varying α and $c_0 \geq 0$; total (P22.98) PROVED for constant α on the start piece and on every graded piece and under the coverage assumption R24-A3c (late truncation (P22.95)); CLAIMED ONLY on ramp pieces; layered cold start OPEN	Section 5.3.7 of the derivation; `sec5_r24_quadrature_start.py`
Joint effective time of a parent and daughter	exists only under N1 and N2; otherwise a two-clock cascade when each species passes R2.1 (E1), and the surrogate or the finite-volume path when a species fails it	PROVED within the class (P22.101); cascade DEMONSTRATED NUMERICALLY; cascade truncation OPEN	Section 5.3.8 of the derivation; `sec5_o11_parent_daughter_tau.py`
Positivity of second-order integrators	no rational order-2 scheme is positive for every step over $\mathcal{S}$; TR-BDF2 for $h\beta_B \leq 1+\sqrt{2}$, CN for $h \leq h^*_{CN}$ (P21.125); exponential integrator for every step	PROVED; per-matrix TR-BDF2 threshold DEMONSTRATED NUMERICALLY; a fixed matrix size $N \geq 3$ OPEN	Section 4.6.1 of the derivation; `sec4_p2_positive_second_order.py`

Property	Statement	Class	Basis
Modal against finite volume agreement	difference falls by a factor of at least 1.8 per halving	DEMONSTRATED NUMERICALLY	V7
Bit-identical regression with v1.0	adding exactly 0.0 leaves every float unchanged	DEMONSTRATED NUMERICALLY in Python; CLAIMED ONLY for Rust until V1 and VT16 run	Section 4.3 detail (ii) of the derivation; VI4
Choice of cumulative yield	adequate when the precursor delay is short	PROVED for a single precursor (Eq. (P21.12)); an approximation otherwise	Section 3.3.2 of the derivation

9.3 Open items

- O1. Capture and burn-out cross sections, the Ag-110m isomeric fraction and flux histories are NOT VERIFIED, as are the daughter diffusivities, the kernel heavy-metal density n_{HM} and the energy per fission. Together these block E5, the burnup route of Eq. (P21.9) and the chains.
- O2. The warning factor of step E1 (default 0.1) has no source; it is a design proposal.
- O3. The tolerances of V7 and V6a come from the first run of the transcription and should be revisited when the Rust code runs.
- O4. Resolved for a kernel-uniform start and source (2026-11-13): Bound A (P21.106) and Bound B (P21.112) of Section 3.5.3 of the derivation bound the truncation of the modal balance form in the layered particle (PROVED under completeness and the classical min-max and Sturm oscillation theorems, so the result rests on O14; DEMONSTRATED NUMERICALLY; both conservative). For $S = 0$ and an absorbing surface, with or without decay and in effective time, the computable tail bounds (P22.90) and (P22.91) of Section 5.3.7 also apply; the start bound (P22.93) is PROVED for constant α and holds in effective time only with α_0 replaced by α_- and each E_n integrated exactly; the late truncation bound (P22.95) needs the coverage assumption R24-A3c. Still OPEN: a bound that uses the cancellation between weights of opposite sign, and a bound after a projection at a temperature switch other than the general form of Bound B.
- O5. The gas constant and the decay treatment used by the CRP-6 participants remain NOT VERIFIED. Cases V11 and V12 were run in the transcription with $R_g = 8.3145$ (the value printed in IAEA, 2012, Eq. (10.5a)) and no decay. Step C2 offers the SI value and 8.3143 only; with either one both cases stay inside their tolerances (B7 review), so a third option is needed only if the review wants the printed TECDOC-1674 constant itself.
- O6. Resolved in the B7 review. Section 4 of the derivation referred to the v1.0 step at "lines 171 to 180", which is the assembly loop inside `backward_euler_step` (lines 162 to 182 in the tag, changed lines 178 and 179). The table of Section 4.3 now names both ranges.
- O7. The conversion between the fluence cutoffs $E > 0.18$ MeV and $E > 16$ fJ is NOT VERIFIED. Step C1.4 refuses fluence input with any other cutoff tag.
- O8. The ramp shape of the CRP-6 lists is assumed linear, and row 8a must be re-read from the page image.
- O9. As-run AGR-1 furnace histories are not in the corpus. The 30 °C start of the reconstructed plan is an assumption supported only by the printed total durations.

- O10. Updated 2026-11-13 (Section 5.4.6 of the derivation). Staircase: order 1 uniform in the stiffness PROVED; order 2 PROVED for shared scaling and, up to a factor $1 + \ln(T/h)$ on a fixed mesh with a mesh-dependent constant, for layer-specific Q (Theorem D). OPEN: an order above 1 uniform in the mesh stiffness for layer-specific Q (observed 1.75 in the test case and 1.51 to 1.96 in a second synthetic case; 7/4 conjectured as the limit, CLAIMED ONLY).
- O11. Resolved in part on 2026-11-13 (Section 5.3.8 of the derivation). Theorem 1 gives necessary and sufficient conditions for a joint effective time (N1 and N2; PROVED in the class (P22.101)); equal activation energies are necessary and not sufficient, and no linear ramp with $\theta = Q/R_g \neq 0$ qualifies when $b\lambda_P \neq 0$. Proposition 2 gives an exact two-clock cascade whenever each species passes R2.1. Still open: a truncation bound for the cascade double series; transformations outside the class; data for daughter diffusivities (O1). The steps that would use these results are proposed in Section 9.5; until then the finite-volume path is the only route.
- O12. Resolved for shared scaling on 2026-11-13 by Section 5.2.5 of the derivation: option A by (P22.69) to (P22.72), option E by (P22.73), (P22.74) with ζ_E; the present F3 value with Λ_1 at the highest temperature is a PROVED sufficient cap in that case for option A (for option E the lag L_1 enters and ε_1 is not proved for that value). For layer-specific activation energies the rule bounds the frozen-coefficient part by ε_1 ((P22.78); option A for any history, option E on heating only); the non-adiabatic remainder R_{na} is CLAIMED ONLY to be $O(\varepsilon_D)$, supported numerically on one history, and has no proved bound. With decay the release-fraction error is $O(\zeta)$ with no sharp a priori bound. A source term is not covered.
- O13. Resolved. The values of N_A and k_B were read in the SI Brochure (BIPM, 2019, Section 2.2 and Table 1, p. 124, version 4.01 of June 2026).
- O14. Partly resolved. The spectral theorem used for completeness was read in Melrose (2009, Lecture 20, Theorem 15, p. 116). A citation for the compactness of an operator with a square-integrable kernel is still NOT VERIFIED (B3 item 2). O14 now also carries the truncation bounds of O4.
- O15. The helper relations (H21.1) to (H21.3) and (H22.1), (H22.2) are not rows of the B3 reconciliation. (H21.x) were proved in the check script and re-derived by substitution here; (H22.1) is a design guard; (H22.2) is a sufficient structural test.
- O16. New 2026-11-13 (Section 4.6.1 of the derivation). No second-order integrator is chosen (decision 15). OPEN: positivity of an order-2 scheme over all matrices of one fixed size $N \geq 3$, which matters because one run has a fixed mesh; for the frozen-matrix exponential integrator with $D(T(t))$, which is the staircase of block R4, the part of its stiff-regime order left open by Section 5.4.6, namely an order above 1 uniform in the mesh stiffness for layer-specific Q (O10); the variable-step BDF2 closure; second-order stages with $D(T(t))$.
- O17. New 2026-11-13 (Section 5.3.7 of the derivation). Start of the release quadrature in R2.4 step 3: a layered particle with a long cold start (large λt_c) has no proved bound (OPEN); the graded Gauss-Legendre bound (P22.97) on ramp pieces is CLAIMED ONLY; a Robin surface and a source are not treated.

9.4 Decisions requested before any Rust is written

1. **Decision 1, source region.** Allow a single-region particle in `Config::validate`, or add a per-region source flag (default: kernel only)? The Booth cases V6a, V6b

and V19 need one of the two.

2. **Decision 2, input of the source.** Take the fission rate density directly (the plan above), or also accept burnup or power? The second option needs n_{HM} or the energy per fission, which are not in the sources (O1).
3. **Decision 3, step average.** Option A (exact step average, recommended in Section 4.3 of the derivation) as the only mode, or keep option R for comparison with older runs?
4. **Decision 4, outside inventory.** Report the Backward Euler M_{out} of Eq. (P21.73) only, or also the exponential form of Eq. (P21.74)?
5. **Decision 5, warning threshold.** Keep the proposed default of 0.1 for the precursor warning, or choose another rule?
6. **Decision 6, step limits.** Make the decay limit (P21.77) and the diffusion limit (P21.78) hard caps on the step, or only warnings with the user step respected?
7. **Decision 7, scope of block R1.** Keep the modal block restricted to a kernel-uniform source and start profile for v2.0, with the general projection as FUTURE?
8. **Decision 8, conductance time level.** The task brief says conductances at $T(t^{n+1})$ (option E). The approved text recommends option A. Both are specified; the default here is A. Which one should be the default?
9. **Decision 9, jump representation.** Two history kinds (linear with strictly increasing times, and constant intervals) or one kind that allows repeated times?
10. **Decision 10, planner.** Accept the merged planner of block F: one `growth` (default 2.0), restart at `dt_first` after a source switch or a temperature jump, landing by halving the remainder, $\varepsilon_D = 0.1$, the factor 2 in the guard, and a FIXED mode that reproduces the v1.0 grid?
11. **Decision 11, gas constant.** SI default with a `FIXED_8_3143` option (as proposed), or a per-row value from the register?
12. **Decision 12, access to eigenfunctions.** Block R3 needs the layer states of the shooting solution, which are private in `sturm_liouville.rs`. Add public accessors to that file, or copy the shooting code into the new module?
13. **Decision 13, reference temperature of the effective time.** The highest temperature of the history (proposed) or the temperature at the end of the main hold?
14. **Decision 14, staircase code.** Keep block R4 in the library behind a test-only switch, or only as a test helper?
15. **Decision 15, integrator (added 2026-11-13).** Backward Euler only, Backward Euler with TR-BDF2 as an option (with or without the positivity cap $h\beta_B \leq 1 + \sqrt{2}$), or Backward Euler with the exponential integrator for constant-temperature segments (Section 4.6.1 of the derivation)?
16. **Decision 16, proposals of Section 9.5 (added 2026-11-13).** Which of the proposed step changes of Section 9.5 should enter the pseudocode before any Rust is written?

The work stops here for the user's review. No Rust or test file of the repository has been touched. The other new repository files are documentation, data and Python only: the check scripts of Section 10 and `scripts/derivation_checks/`, the parameter register under `data/` with its scripts in `scripts/data_register/`, and the change list `docs/development/v1_to_v2_changes.md`.

9.5 Proposed step changes from the open-item work (2026-11-13, not applied)

The open-item work of 2026-11-13 (Sections 3.5.3, 4.6.1, 5.2.5, 5.3.7, 5.3.8 and 5.4.6 of the derivation) suggests the step changes below. None of them is applied to the blocks above: every verified result left the existing steps valid, so no algorithm was

changed, and only notes, status rows and open items were updated. The two notes that corrected a statement are the F3 note, which now records that the existing value is a proved sufficient cap for shared scaling with option A, and the R2.4 step 3 note, whose remark that the balance form avoids the start problem was wrong ((P22.81)). A third correction is a comment only: the order annotations of the MIDPOINT and AVERAGE lines in step R4.1 now read “order 2 for F; on a fixed mesh up to 1+ln(T/h)” and “order 2 as MIDPOINT” in place of “order 2”, and the steps themselves are unchanged. Each proposal below is DERIVED NOT IMPLEMENTED and waits for decision 16.

Block R1 (from O4, Section 3.5.3).

- New step R1.0, mode count, before R1.1: from the radii and diffusivities compute τ, a, c, J and Λ^*; with $K =$ `gen_max + decay_constant * (inv0 + gen_total_max)` and the user tolerance choose `n_modes` by (P21.110), which needs $N > J$ and $\ell_{N+1} > \Lambda^*$, together with the start-inventory condition $C_w(\ell_{N+1}) M_0 X_N(t_1) \le \varepsilon$ at the first output time t_1; pass `lambda_max` $\ge \min\left(D_{\max}(N\pi/R)^2,\ a(N+J)^2\right)$ to `ModalRelease::new` so that the first N modes are all included. The condition $\ell_{N+1} > \Lambda^*$ can force a large N (2020 modes in one verification case).
- Step R1.5, diagnostics: add `trunc_bound_rel` from (P21.106) at each output time, with X_N from (P21.105) and $\Lambda_{\text{cut}} =$ `lambda_covered`, and `trunc_bound_FB`, `trunc_bound_RB` from (P21.107) and (P21.108). Bound B (P21.112) needs α_n and β_n separately and stays FUTURE while v1.0 keeps only w_n.
- Step R3.5: note that Bound A does not apply after a switch; the general Bound B is the FUTURE route there.

Block F (from O12, Section 5.2.5, and P2, Section 4.6.1).

- Step F3: compute $\Theta_{\text{end}} = \int_0^{t_{\text{end}}} (\Lambda_1 + \lambda)\, dt$ (shared scaling: $\Lambda_1^{\text{ref}} \tau(t_{\text{end}}) + \lambda t_{\text{end}}$ from (P22.22); otherwise composite Gauss-Legendre over each history segment with $\Lambda_1(T)$ from the tridiagonal eigenvalue problem or with the upper bound $\hat{\Lambda}_1$), then $\zeta = 2\varepsilon_1 / \Theta_{\text{end}}$ for option A or ζ_E of (P22.74) for option E, citing (P22.69). For constant T this gives the present `dt_diff`, and grid mode FIXED leaves the block before F3, so cases V1 and VT16 are unchanged.
- Step F4: take `dt_diff` out of `dt_cap`, because it changes from step to step, and add the estimate $\Theta_{\text{end}}^2 / (2\varepsilon_1)$ of (P22.71) to the `step_limit` test.
- Step F5: a new sub-step after F5.3 computes `dt_diff` by (P22.72) with two evaluations of Λ_1; in F5.4 the step becomes the minimum of `dt_cap`, `grow_cap`, `dt_T` and `dt_diff`. Keep `dt_T` of (P22.20) for layer-specific Q, since it controls the remainder R_{na}. With two nuclides on the finite-volume path, ζ is the minimum over the nuclides.
- If a second-order integrator is chosen (decision 15): F3 uses (P21.142) with the error constant c_r of that scheme (and the BDF2 start term); F4 gains an optional positivity cap $h \le M_r / \beta_B$ ($M_r = 2$ for CN, $1 + \sqrt{2}$ for TR-BDF2) or $h \le h^*_{CN}$ from (P21.125), hard cap or warning as in decision 6.

Blocks H and J (from P2, Section 4.6.1; only if decision 15 adds an integrator).

- New step H4b: reuse the subtraction-free elimination of H4.3 with $\tau = h/2$ (CN) or $\tau = \beta h$ (TR-BDF2, two solves with one matrix), or the exponential step (P21.137). The property line of H4.4 holds for H4b only in the exponential case, and there only with (P21.139).

- Steps J1.3 to J1.5: the increments of M_{dec}, M_{rel} and the outside inventory use the quadrature of the chosen scheme: trapezoid (P21.126), stage weights (P21.134), modified inventory (P21.130) or exact integrals (P21.140). Step J1.6: the exact test "minimum below 0.0 is a failure" holds only for BE and the uniformised exponential; other schemes need a tolerance or the step cap.

Block R2 (from R24, Section 5.3.7, and O11, Section 5.3.8).

- R2.4 step 3, sub-steps: (3a) compute M_{rel} by (P22.82); the balance form is the same number by (P22.81) and is not a separate option. (3b) Start piece: for a bare kernel with a uniform start the image form (P22.86) or (P22.87) on $[0, t_0]$ with $t_0 = \min(t_B, t)$ (the 16-point quadrature of (P22.87) on a ramp has no proved error bound); otherwise the modal start (P22.83) with $t_0 = \min(t_1, t)$ and the bracket (P22.94) (needs $c_0 \geq 0$) when α varies on $[0, t_0]$. (3c) Late pieces: breakpoints refined to $\tau(b)/\tau(a) \leq 2$, 8-point Gauss-Legendre, bound (P22.97) (PROVED for constant α per piece, CLAIMED ONLY on ramp pieces). (3d) Mode count and t_0 by the rule of Step 7 of Section 5.3.7, with $\bar{W}_N$, $\bar{V}_N$ from (P22.90), (P22.91). (3e) Return the bound (P22.98) with the value; warn when it exceeds the tolerance, and warn when the cold-start term $\lambda t_0 \bar{W}_N$ dominates in a layered particle (O17). Step 5 stays as it is for the reported times; the start piece no longer needs the series to be valid on $[0, t_0]$. All of this assumes an absorbing surface and no source.
- R2.1 chain mode `check_chain(laws_P, laws_D, history)`: run R2.1 on each species and return `Identical` when (H22.2) holds across the two species with ratio 1 in every layer, `SharedClock` when it holds with another ratio (a closed form then exists for $b\lambda_P = 0$ by two copies of (P22.27), on isothermal intervals by (P21.21) and on histories that obey (P22.106) by (P22.108); by Corollary 2 no linear ramp with $Q \neq 0$ qualifies when $b\lambda_P \neq 0$), `TwoClocks` when each species passes alone, and `NotExact` otherwise.
- R2.4 step 6 for `Identical`: $M_P = e^{-\lambda_P t}\hat{M}_P(\tau)$ and $M_D = e^{-\lambda_D t}\hat{M}_D(\tau) + b\lambda_P \frac{e^{-\lambda_P t} - e^{-\lambda_D t}}{\lambda_D - \lambda_P}\hat{M}_P(\tau)$, both $\hat{M}$ from the v1.0 series at the same τ ((P22.109)).
- New block R2.5, two-clock cascade: eigenpairs of both species at their own reference temperature; overlap O by (P21.19), once; τ_P and τ_D by R2.2 and R2.3; the double integrals J_{nm} by Gauss-Legendre on the pieces between breakpoints, graded towards both ends of each piece; inventory by (P22.114) and release by the daughter balance of Section 5.3.8. Its truncation rule is OPEN (O11). Routing: identical diffusivities with E1 for the pair go to R2.4 step 6; each species passing R2.1 goes to R2.5; otherwise the finite-volume path, with the surrogate of Theorem 3 ((P22.118) to (P22.120)) as a diagnostic only.

Block R4 (from O10, Section 5.4.6).

- New sub-step R4.0: place interval ends on every kink and jump of $T(t)$ (Theorem B: exact for piecewise-constant histories, order 1 for a misaligned jump).
- Step R4.2: measure the observed order on a smooth functional such as F; as in the R4 note, a V-norm acceptance test requires an order of at least 1 only, and order 2 of MIDPOINT and AVERAGE is tested on F (an order above 1 in the V-norm rests on numerical evidence only, and its uniform proof is OPEN, O10). Table 5.6, row (P22.54), (P22.55): add a pointer to (P22.139) for the interface defect. An optional case VT18b can test (P22.130).
- With the exponential integrator of decision 15, the frozen-matrix exponential on

the finite-volume matrix is itself a positive staircase; whether R4 then becomes a production path is part of decision 15.

10 Evidence

All commands use `.venv/Scripts/python.exe -I -X utf8`.

Table 10.1 Check scripts.

Script	Folder	Result when this page was written
`b4_phase2_pseudocode_integration.py`	`scripts/derivation_checks/`	32 of 32 PASS
`b4_decay_source_pseudocode.py`	`scripts/derivation_checks/`	55 of 55 PASS (re-run)
`b4_arrhenius_checks.py`	working folder, not in the repository	41 of 41 PASS (re-run)
`sec5_arrhenius.py`, `sec5_effective_time.py`, `sec5_fv_shared_alpha.py`, `sec5_projection_step.py`, `sec5_staircase.py`, `sec5_schedules.py`	`scripts/derivation_checks/`	25, 16, 10, 9, 9 and 17 checks PASS (B3 re-run, not repeated here)
`sec6_pseudocode_checks.py`	`scripts/derivation_checks/`	30 of 30 PASS in the part draft; not reviewed line by line
`sec3_o4_layered_truncation_bound.py`, `sec4_p2_positive_second_order.py`, `sec5_o12_variable_D_step_limit.py`, `sec5_r24_quadrature_start.py`, `sec5_o11_parent_daughter_tau.py`, `sec5_o10_stiff_staircase.py` (open items, 2026-11-13)	`scripts/derivation_checks/`	91, 41, 38, 25, 24 and 22 PASS, 0 FAIL, each ending with ALL PASS (run of 2026-11-13; output in Appendix A.2 of the derivation)

The new script is a Python transcription of the merged planner, of blocks H and J, and of the v1.0 run loop. It does not import the other scripts and writes no file. It checks:

- the planner landing on every event, the cap rules and the restart after a switch (VI1);
- the closure inside and outside, positivity and the generated amount for a joined run, and the failure of the closure when the first-step G_R is used (VI2, VI3);
- bit identity with the v1.0 run loop for constant data (VI4);
- the left-limit rule at a temperature jump and the failure of the naive rule (VI5);
- the difference between the geometric grid with growth 1 and the v1.0 constant step (finding X2);
- option A against Backward Euler in the effective time (VI6);
- the structural test of the effective-time shortcut on three inputs;
- the closed form with E_2 against adaptive quadrature at a change of 16 in $\ln D$ (relative difference 2.3×10^{-15});
- the numbers quoted here: the four decay constants, the Cs-137 source, the two step limits of block F, the scalar example, the five t_{99} values and the three temperature step bounds.

Key output lines of the new script:

```
PASS  every event time is the end of a step by float equality  8 of 8
PASS  joined run, option A: inside closure below 1e-12  1.31e-15
PASS  negative test: booking with the first-step surface conductance breaks the closure (above 1e-8)
       ↪  1.22e-05
PASS  v2 loop equals the v1.0 transcription bit for bit over every step  200 steps, dt 2000.000000 s
PASS  inventory after the run: option A against BE in tau agrees to 1e-11  3.95e-16
```

```
PASS  naive rule is a trap: it produces steps below 1 ms next to the jump
SUMMARY: 32/32 PASS
```

What the scripts do not establish: they check transcriptions in Python floating point, so bit identity (V1, VT16, VI4), the behaviour of the Rust compiler and the speed of the Rust code stay open until Rust tests exist. Case V8 and the transient Booth tolerance of Section 4.8 test 6 were not run. The joined scheme has not been convergence-tested.

References

Basu, S. K., & McCutchan, E. A. (2020). Nuclear data sheets for A = 90. *Nuclear Data Sheets, 165*, 1-329. https://doi.org/10.1016/j.nds.2020.04.001

Browne, E., & Tuli, J. K. (2007). Nuclear data sheets for A = 137. *Nuclear Data Sheets, 108*, 2173-2318. https://doi.org/10.1016/j.nds.2007.09.002

Bureau International des Poids et Mesures. (2019). *The International System of Units (SI)* (9th ed., version 4.01, June 2026). BIPM. https://doi.org/10.59161/AUEZ1291 (Section 2.2, pp. 123-124, and Table 1, p. 124; kelvin, p. 129, and mole, p. 130, both in Section 2.3.1; English text PDF opened from https://www.bipm.org/en/publications/si-brochure)

Gürdal, G., & Kondev, F. G. (2012). Nuclear data sheets for A = 110. *Nuclear Data Sheets, 113*, 1315-1561. https://doi.org/10.1016/j.nds.2012.05.002

idaholab. (2026). *moose* [Computer software]. GitHub. https://github.com/idaholab/moose (`force_step_every_function_point`, via audit 08)

International Atomic Energy Agency. (1997). *Fuel performance and fission product behaviour in gas cooled reactors* (IAEA-TECDOC-978). IAEA. (Tables A-1 to A-3, the source of the Cs diffusion correlations in the register; URL NOT VERIFIED.)

International Atomic Energy Agency. (2012). *Advances in high temperature gas cooled reactor fuel technology* (IAEA-TECDOC-CD-1674) [CD-ROM]. IAEA. (Eqs. 7.1 to 7.8, printed pp. 317-319; Eqs. 10.1 to 10.7, printed pp. 529-544; Table 8.1; URL NOT VERIFIED.)

International Atomic Energy Agency, Nuclear Data Section. (n.d.). *LiveChart of Nuclides, data download API v1* [Data set]. Retrieved November 5, 2026, from https://www-nds.iaea.org/relnsd/vcharthtml/api_v0_guide.html (source of the ENSDF half-lives and of the JEFF-3.1.1 fission yields quoted here)

Melrose, R. (2009). *Lecture notes for 18.102 Introduction to functional analysis, Spring 2009* (Lecture 20, Theorem 15, p. 116) [Course notes]. MIT OpenCourseWare. https://ocw.mit.edu/courses/18-102-introduction-to-functional-analysis-spring-2009/ (instructor named on the course page; the lecture PDF prints no author)

STRATO project. (2026). *STRATO v1.0.0* [Computer software]. https://github.com/weewzr/STRATO/tree/v1.0.0 (files `src/numerical/finite_volume.rs`, `src/analytical/sturm_liouville.rs` and `src/config.rs`, quoted by line number).

The equations (P21.1) to (P21.142) and (P22.1) to (P22.142) are those of Sections 3 to 5 of the derivation (`docs/mathematics/STRATO_Phase2_Derivations.md`) and carry their own references there; (P21.93) to (P21.142) and (P22.60) to (P22.142) were added on 2026-11-13.